Manual Calibration
Complete Technical Report

Force estimation, contact detection and nine-taxel localization
A self-contained source document for thesis writing

Prepared: 18 October 2026
Model experiments: 9 September 2026
Dataset: Manual Calibration (2).zip only

Best evaluated results

Table 1. Recommended headline models and held-out metrics

Task	Best model	Primary result	Supporting results
Localization	Tuned Logistic regression	84.88% accuracy	85.00% macro F1
Contact detection	Tuned RBF SVC	99.24% accuracy	99.54% balanced accuracy; 99.09% recall
Force estimation	Tuned ensemble	0.4972 N MAE	0.6299 N RMSE; R^2 = 0.3113

The two classifiers have the highest observed recording-weighted accuracy among the evaluated models. The force ensemble has the lowest observed recording-weighted mean absolute error. Each task uses one winning model for all its headline metrics. [S1–S3]

Predictions were generated with six complete TEST groups held out in turn. Force and localization cover 7,154 eligible frames from 54 recordings; contact detection covers 8,506 frames from 65 recordings. All reported model metrics give each recording equal total weight.

Model-family ranking is retrospective on these outer cross-validation results. These are held-out fold predictions, not an untouched prospective test of a final full-data model. The report preserves all 50 evaluated comparison rows and all 178 tuning configurations.

This file contains the complete methodology, exact segmentation, winning fold configurations, full model comparisons, separately labeled normalized and raw-count confusion matrices, spatial and optical heatmaps, limitations, reproducibility details, and a ready-to-use thesis-writing prompt. Existing reports and competing model artifacts remain intact.

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Percentages in metric and normalized-matrix tables are displayed on a 0–100 scale. Force errors are in newtons. Raw confusion matrices contain integer frames. Internal source IDs [S1–S16] resolve in the evidence register at the end.

1. Purpose, system and dataset

The study estimates applied force from optical measurements, detects contact, and identifies the contacted taxel in a fixed 3 × 3 array. It compares model families under a common grouped validation design and evaluates improvements from richer optical features, hyperparameter search, causal history, threshold selection and ensembles.

The SPARE application records a camera stream alongside an Arduino Nano/HX711 reference-force stream, unloaded optical baselines, timestamps and nine regions of interest (ROIs). Reference force provides supervision and evaluation targets; it is absent from optical model inputs. The separate printer-driven automatic-calibration path is excluded. Historical hardware records identify an Arducam IMX179 module, but do not establish identical camera controls or a traceable force-uncertainty budget for every archived recording. [S4, S5, S15]

Table 2. Task definitions and evaluation population

Task	Target	Evaluation population
Force estimation	Continuous reference force, N	Eligible manual press frames
Localization	Discrete taxel class T1–T9	Same eligible manual press frames
Contact detection	0 = dedicated no contact; 1 = eligible press	Eligible press plus dedicated no-contact frames

Table 3. Manual archive composition

Role	Recordings	Frames	Analysis use
Primary press	54	24,453	Eligible frames supply all positive tasks
Dedicated no contact	11	1,352	Contact negatives; training-only optical floors
Replay only	18	3,513	Excluded from fitting and scoring
Total	83	29,318	Manual archive only

Six TEST groups each contain one primary recording per taxel. Dedicated no-contact recordings remain whole and are assigned two each to TEST1–TEST5 and one to TEST6. The preserved feature store is feature-store-fa2500440adab6ca. Archive and partition provenance were checked in the training audit. [S4–S6]

Table 4. Primary-frame eligibility after fold-specific force alignment

Category	Frames
All primary frames	24,453
Unpaired with valid reference force	142
Aligned force below 0.05 N	723
Aligned force above 3 N	16,434
Eligible press frames, inclusive 0.05–3 N	7,154

Frames outside the eligible press interval are excluded, not relabeled as no contact. The 0.05–3 N interval is an exploratory analysis restriction; it is not an established detection limit or validated full operating range. Most primary frames are outside this interval.

2. Exact image segmentation and ROI geometry

The quantitative image source is session_video.mp4. Overlay, processed and motion-magnified videos are excluded. A raw 640 × 480 frame is rotated 90° clockwise and then mirrored horizontally, producing a 480 × 640 oriented frame. Already oriented frames are not transformed again. Coordinates below are oriented camera pixels; the origin is the upper-left corner. [S6, S8]

Table 5. Fixed taxel rectangles; ROI layout roi-9fbc67c50ee3bc7145fa

Taxel	Row	Col.	x	y	Width	Height	Area (px²)
1	1	1	98	161	82	89	7298
2	1	2	197	156	76	99	7524
3	1	3	312	154	58	97	5626
4	2	1	100	268	73	89	6497
5	2	2	204	264	76	96	7296
6	2	3	314	260	66	94	6204
7	3	1	104	383	73	77	5621
8	3	2	194	394	87	66	5742
9	3	3	312	379	61	77	4697

Rectangles use the array slice image[y:y+height, x:x+width]. Taxels T1–T3 are the first row, T4–T6 the second, and T7–T9 the third. These are fixed manually configured regions, not learned segmentation masks.

Table 6. Pixel operations and their exact settings

Operation	Definition / setting
Color conversion	BGR to OpenCV HSV; use the 8-bit V channel
Unloaded baseline B(x,y)	Per-recording saved per-pixel median V from baseline_data.npz
Signed change	$\Delta V(x,y,t) = V(x,y,t) - B(x,y)$
Positive change	$\Delta V^+(x,y,t) = \max(\Delta V(x,y,t), 0)$, applied per pixel
Active mask	$\Delta V^+ \geq 10.0$ camera V levels
Active fraction	Active-pixel count divided by fixed ROI area
Centroid	Intensity centroid weighted by all ΔV^+ values; not only active-mask pixels
Hue-only saturation setting	Minimum saturation = 10.0; this is not the V activity threshold
Morphology / learned segmentation	None / none

Per-ROI measurements retain positive and signed light sums, active fractions, signed median and median absolute deviation (MAD), raw V mean/median/maximum and centroid coordinates. Signed features preserve intensity decreases. A missing centroid is represented by the normalized ROI center (0.5, 0.5) and a separate absence flag.

Optical activity is an image statistic. It does not establish physical contact area, calibrated illuminance or the anatomical/mechanical boundary of contact.

3. Feature construction and preprocessing

Training-only normalization

For each taxel, the fitted positive-light floor is the median of the within-recording medians of positive-light sums in training no-contact recordings. The positive feature is max(raw positive sum − floor, 0) / ROI area. Signed light is divided by area without clipping. Floors are refitted at each inner and outer training boundary. [S5, S8, S9]

Weighted StandardScaler uses training rows only. For feature k, $z_k = (x_k - \mu_k) / \sigma_k$, with weighted mean and population variance fitted on training data. Centering is disabled only for the positive-only force pack; variance scaling remains enabled. Zero-variance scale follows the library implementation. The selected best models all use centered and scaled features.

Table 7. Feature packs and exact dimensions

Pack	Dimensions	Contents
positive	9	Floor-corrected positive light per pixel, T1–T9
optical	27	positive + signed light per pixel + active fraction
shape	90	optical plus seven 9-channel normalized/centered pattern blocks
spatial	180	shape plus signed statistics, centroids, presence and intensity transforms
invariant	117	63 pattern channels plus six 9-channel spatial/statistical blocks
temporal	288	spatial plus four 27-channel causal-history/difference blocks

Across the nine taxels, $\text{unit}(x) = x / (\Sigma|x| + 10^{-6})$, and $\text{contrast}(x) = x - \text{median}(x)$. The seven pattern blocks are unit(positive), unit(contrast(signed)), unit(active), contrast(signed), contrast(active), unit(contrast(raw V mean)), and unit(contrast(signed median)).

Spatial adds signed median, signed MAD, normalized centroid x and y, centroid-presence flags, contrast(raw V mean), contrast(raw V median), raw V maximum / (raw V mean + 1), log(1 + positive), and sign(signed)·log(1 + |signed|). Invariant contains the seven pattern blocks plus unit(MAD), centroid x/y, presence, unit(contrast(raw V median)), and the maximum-to-mean ratio. Its name does not prove invariance to all illumination or geometry changes.

Causal temporal history

The optical pack supplies fast and slow exponential moving averages (EMAs), with time constants 0.3 s and 1.0 s. Temporal concatenates spatial, fast, slow, optical − fast, and fast − slow. At positive time step Δt, $\alpha = 1 - \exp(-\Delta t/\tau)$ and $h_t = h_{(t-1)} + \alpha(x_t - h_{(t-1)})$. History restarts on a new recording, nonpositive Δt, or a gap greater than 0.3 s. Output smoothing is a separate selection over τ {0, 0.3, 1.0} s.

Inputs contain no reference force, true taxel, TEST number, session identity, absolute time or recording duration. Identity and relative timestamps only partition/reset history and supply time differences. The exact ordered feature-construction function is printed in Appendix C.

4. Alignment, validation and metric definitions

Reference-force alignment

At optical time t, the target is reference force at t − L. Linear interpolation remains within a recording; duplicate reference timestamps are reduced by their median. Extrapolation and interpolation over gaps greater than 300 ms are prohibited. Lag candidates are 0, 100, 200, 300, 400 and 500 ms. Inner grouped NNLS MAE selects the smallest lag within 0.5% of the best error. TEST1–TEST6 use outer lags 300, 300, 200, 300, 300 and 300 ms, respectively. Tuning reuses these fold-local alignment decisions to retain identical evaluated rows and targets. [S4, S5]

Nested whole-group evaluation

Each outer fold holds out one complete TEST group and its assigned no-contact recordings. The other five groups form five inner folds: four groups fit, one validates. Preprocessing, hyperparameters, feature pack, history, correction, thresholds and model choice are determined without the outer group. Fifteen possible four-group fits are cached; only the relevant inner-validation group is consumed for an outer selection. Thirty inner train/validation boundaries were audited. The final research refit uses all 15 leave-two-groups-out splits for selection, then fits the full pool. [S5, S9]

Earlier optical rows from a held-out recording can initialize causal history, even if their force lies outside the scored interval. Their labels are not consumed. These are chronological stream predictions at stationary per-recording target taxels, not independent cold-start frames or demonstrated moving-contact localization.

One weighting convention for reported results

For an evaluated recording s with n_s eligible rows, each row receives weight $w_i = 1/n_s$. Each recording therefore contributes one total unit. All headline, comparison and confusion-matrix percentages use these weights. For a force-bin subgroup, n_s is recomputed using only that bin. Raw matrix counts are deliberately unweighted integers. Training may use additional class/bin balancing; that is separate from evaluation weighting.

Table 8. Metric definitions; e = predicted force − reference force

Metric	Definition
MAE; RMSE; bias	$\Sigma w\|e\|/\Sigma w$; $\mathrm{sqrt}(\Sigma we^2/\Sigma w)$; $\Sigma we/\Sigma w$
R^2	$1 - \Sigma we^2 / \Sigma w(y - \text{weighted mean of } y)^2$
Accuracy	Weighted correct predictions / total weight
Recall; precision	TP/(TP+FN); TP/(TP+FP), using weighted outcomes
Specificity; FPR	TN/(TN+FP); FP/(TN+FP)
Balanced accuracy	Mean of class recalls; contact: (recall + specificity)/2
Class F1; macro F1	2TP/(2TP+FP+FN); arithmetic mean of class F1 scores
Row-normalized confusion matrix	100 × weighted cell / weighted actual-class row total

Pooled macro F1 need not equal the mean of six fold macro F1 values. Nine-class accuracy equals balanced accuracy here because each taxel has six equally weighted recordings. Contact accuracy and balanced accuracy differ because there are 54 positive and 11 negative recordings. No new confidence interval or significance claim is inferred from thousands of temporally correlated frames.

5. Models, search and improvement strategy

The fixed benchmark evaluates nine force regressors and six classifiers for each classification task. Seed 1729, 22 training subsets and 21 task-specific estimators yield 462 fits. Force uses the positive 9-feature pack; classifiers use optical 27. Force fitting balances recordings and 0.25 N bins, localization balances recordings, and contact balances classes and recordings. Fixed models use native decision rules without temporal smoothing, threshold tuning or force-output clipping. [S4, S7]

The tuning run uses seed 1741 and a finite grid of 178 estimator/feature configurations: 62 force, 64 localization and 52 contact. Fifteen four-group training subsets yield 2,670 inner estimator fits. Additional output-history, correction, threshold and ensemble variants reuse cached inner predictions and are not counted as additional grid candidates. Appendix A lists every candidate, not only the winning settings. [S5, S8]

Table 9. Search mechanisms and exact decision rules

Mechanism	Rule
Estimator and feature search	Candidate grids in Appendix A; common factory settings in Appendix B
Output history	τ = 0, 0.3 or 1.0 s; index h = 0, 1 or 2
Force output	Clip raw prediction to ≥0, apply selected output EMA, add inner-OOF residual offset, clip to ≥0 again
Force offsets	None (=0), recording-weighted mean residual, or weighted median residual; select by inner MAE
Localization	Smooth per-class scores, then argmax in class order T1–T9; select internally by macro F1
Contact score	For SVC: sigmoid(decision_function); otherwise class-1 predict_proba; then selected output EMA
Contact threshold	Exact weighted ROC operating points from inner OOF scores; balanced policy maximizes balanced accuracy
Alternative contact policy	Maximize inner recall subject to inner FPR ≤5%; select family by recall, then FPR
Ensembles	Equal mean of three best distinct tuned families; each member has its own selected history/correction
Training weights	Recording weights, plus class balancing for contact; force for_000 alone uses bin weights in the tuning grid

The tuned selector chooses across families using inner MAE, localization macro F1 or contact balanced accuracy; it adopts an ensemble only if the inner criterion improves. The tuned family rows instead restrict that inner search to one family. The present headline ranking follows the requested highest classification accuracy and lowest force MAE across the evaluated outer results.

SVC sigmoid/softmax transformations provide score representations, not calibrated probabilities. A 1.0 s EMA time constant does not prove a measured 1.0 s event delay. The FPR ≤5% condition is an inner-selection constraint, not a guaranteed outer FPR.

6. Complete model comparisons

All 50 saved comparison rows are retained below: 19 force, 15 localization and 16 contact. Rows are ordered by the requested criterion. “Tuned” identifies the expanded search, while unprefixed family names identify the fixed benchmark. “Nested selected” and “Tuned selected” are evaluated selection procedures, not new estimator families. [S2, S3]

6.1 Force estimation — rank by lowest MAE

Table 10. All force models on 7,154 frames / 54 recordings

Rank	Model / procedure	MAE (N)	RMSE (N)	Bias (N)	R^2
1	Tuned ensemble	0.4972	0.6299	0.0827	0.3113
2	Tuned selected	0.4991	0.6323	0.0784	0.3061
3	Tuned HistGB	0.5030	0.6374	0.0739	0.2949
4	Tuned XGBoost	0.5035	0.6404	0.0924	0.2883
5	Tuned Extra Trees	0.5046	0.6359	0.0819	0.2982
6	Tuned RBF SVR	0.5135	0.6397	0.0662	0.2899
7	Tuned Random Forest	0.5221	0.6606	0.1007	0.2427
8	Tuned frame-only	0.5393	0.6827	0.1118	0.1910
9	Tuned MLP	0.5431	0.6884	0.1207	0.1775
10	Extra Trees	0.6372	0.7705	-0.2894	-0.0303
11	Nested selected	0.6372	0.7705	-0.2894	-0.0303
12	RBF SVR	0.7015	0.8617	-0.3291	-0.2885
13	Monotonic HistGB	0.7084	0.8449	-0.3268	-0.2388
14	Monotonic XGBoost	0.7159	0.8457	-0.3400	-0.2412
15	Isotonic	0.7233	0.8487	-0.3801	-0.2500
16	Median baseline	0.7536	0.8633	-0.4111	-0.2935
17	Positive Ridge	1.5292	1.9017	-0.9075	-5.2762
18	NNLS	1.5295	1.9027	-0.9063	-5.2829
19	Total light linear	1.5614	1.8776	-0.9492	-5.1186

The tuned ensemble reduces MAE from the strongest fixed model’s 0.6372 N to 0.4972 N: a 0.1400 N absolute reduction and 21.97% relative reduction. RMSE falls from 0.7705 N to 0.6299 N. Bias changes from −0.2894 N to +0.0827 N, and R^2 rises from −0.0303 to 0.3113. These describe the overall evaluation; low-force error remains substantial.

No classification accuracy is assigned to a continuous force regressor. Negative fixed-model R^2 values mean worse squared error than the evaluation-set weighted mean reference. They do not mean negative physical accuracy.

6.2 Localization — rank by highest accuracy

Table 11. All localization models on 7,154 frames / 54 recordings

Rank	Model / procedure	Accuracy (%)	Balanced accuracy (%)	Macro F1 (%)
1	Tuned Logistic regression	84.88	84.88	85.00
2	Tuned ensemble	84.71	84.71	85.12
3	Tuned selected	84.40	84.40	84.75
4	Tuned frame-only	82.84	82.84	82.68
5	Tuned Extra Trees	81.96	81.96	83.35
6	Tuned HistGB	81.75	81.75	82.94
7	Tuned MLP	81.65	81.65	82.46
8	Tuned RBF SVC	77.80	77.80	78.71
9	RBF SVC	73.87	73.87	73.92
10	Logistic regression	73.79	73.79	74.00
11	HistGB	71.31	71.31	72.33
12	Extra Trees	71.04	71.04	71.70
13	Random Forest	70.66	70.66	71.49
14	XGBoost	70.37	70.37	71.38
15	Nested selected	70.29	70.29	70.14

Tuned Logistic regression leads accuracy at 84.88%, compared with 73.87% for the most accurate fixed family, RBF SVC: an improvement of 11.01 percentage points. Tuned Logistic regression has 85.00% macro F1. The tuned ensemble has a slightly higher macro F1 (85.12%) but slightly lower accuracy (84.71%); Logistic regression is the headline winner under the requested accuracy criterion.

The frame-only result excludes temporal features and output smoothing from its inner selection. It is an evaluated constrained selection procedure; its difference from the main winner cannot be attributed solely to one feature or one smoothing setting because selected models may also differ.

6.3 Contact detection — rank by highest accuracy

Table 12. All contact models on 8,506 frames / 65 recordings

Rank	Model / procedure	Acc. (%)	Bal. acc. (%)	Macro F1 (%)	Recall (%)	Precision (%)	FPR (%)
1	Tuned RBF SVC	99.24	99.54	98.67	99.09	100.00	0.00
2	Tuned XGBoost	99.03	99.41	98.31	98.83	100.00	0.00
3	Tuned Random Forest	98.76	99.17	97.86	98.56	99.95	0.23
4	Tuned FPR <=5%	98.16	95.40	96.64	99.57	98.24	8.76
5	Tuned Extra Trees	97.65	96.04	95.84	98.47	98.70	6.39
6	Tuned selected	97.51	98.50	95.81	97.00	100.00	0.00
7	Tuned Logistic regression	97.41	98.44	95.66	96.89	100.00	0.00
8	Tuned HistGB	97.12	96.12	95.00	97.63	98.89	5.40
9	Tuned frame-only	96.25	97.09	93.78	95.82	99.65	1.64
10	Random Forest	91.91	91.86	87.15	91.93	98.22	8.20
11	HistGB	89.03	90.31	83.52	88.38	98.24	7.76
12	XGBoost	88.52	90.03	82.91	87.74	98.25	7.69
13	Logistic regression	88.28	90.44	82.76	87.17	98.55	6.29
14	RBF SVC	88.20	90.97	82.81	86.78	98.88	4.83
15	Nested selected	87.95	89.54	82.20	87.14	98.15	8.05
16	Extra Trees	87.78	89.77	82.08	86.76	98.33	7.23

Tuned RBF SVC leads at 99.24% accuracy, compared with 91.91% for fixed Random Forest: +7.34 percentage points. Balanced accuracy rises from 91.86% to 99.54% (+7.68 points). Its contact recall is 99.09%, precision is 100.00%, specificity is 100.00%, and FPR is 0.00% on these held-out recordings.

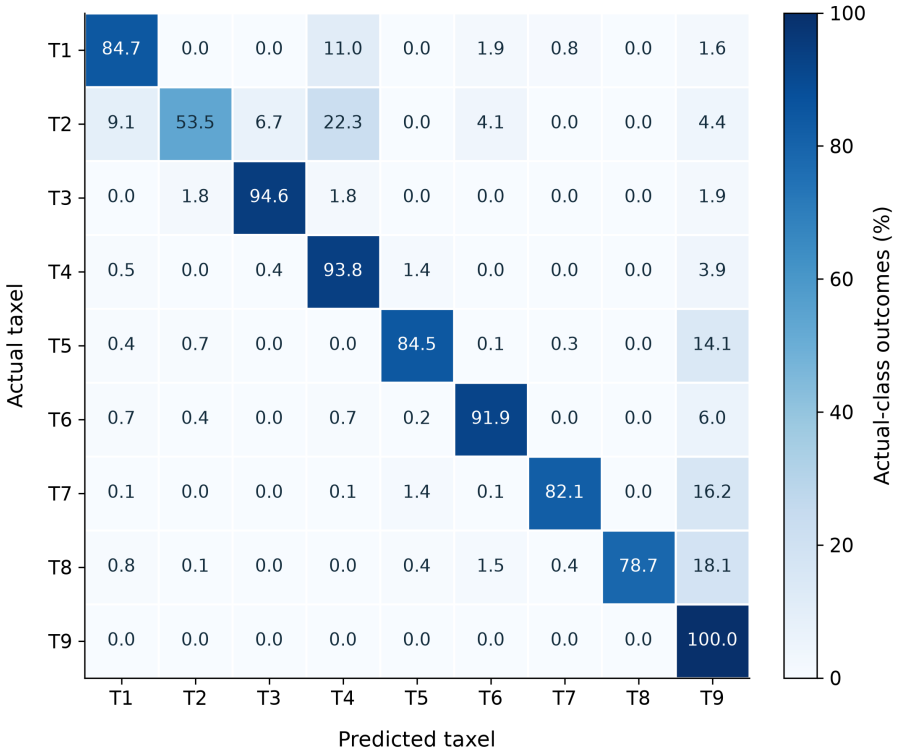
The “Tuned FPR ≤5%” procedure attains 99.57% recall but 8.76% outer FPR. Its name describes the inner constraint; the observed outer value is preserved explicitly. It is not the accuracy winner. Tuned contact AUC is not reported because adaptive fold models and uncalibrated score scales make pooled discrimination scores difficult to interpret consistently. No missing AUC is treated as zero.

7. Localization confusion matrices and spatial performance

Localization — best accuracy model

Tuned Logistic Regression | Accuracy 84.88%

Six held-out TEST groups | 7,154 frames | 54 recordings



ROW-NORMALIZED (%) — each recording has equal total weight

Manual calibration only | Positive force range: 0.05–3 N

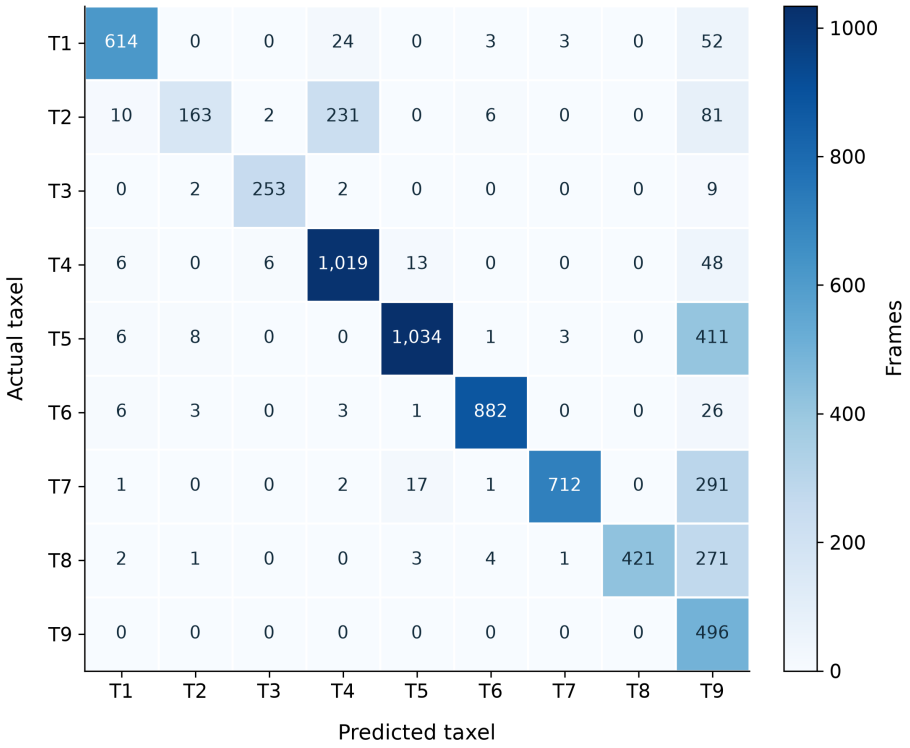
Accuracy uses equal recording weights; raw counts separately show every evaluated frame.

Figure 1. Localization: recording-weighted row-normalized 9 × 9 matrix. Tuned Logistic regression. Actual taxel is the row; predicted taxel is the column. Each frame receives weight 1/n for its recording before row normalization. The diagonal is taxel recall. Six held-out TEST groups, 7,154 frames and 54 recordings. Numeric values appear in Table 13.

Localization — best accuracy model

Tuned Logistic Regression | Accuracy 84.88%

Six held-out TEST groups | 7,154 frames | 54 recordings



RAW COUNTS — unweighted integer frames

Manual calibration only | Positive force range: 0.05–3 N

Accuracy uses equal recording weights; raw counts separately show every evaluated frame.

Figure 2. Localization: separately labeled 9 × 9 raw frame counts. The same winning-model held-out predictions, counted as integers without recording weights. Raw row totals differ because recordings contain unequal numbers of eligible frames. These counts must not be divided to reconstruct the weighted percentages.

Machine-readable localization matrix values

Table 13. Localization row-normalized weighted percentages (%)

Actual / predicted	T1	T2	T3	T4	T5	T6	T7	T8	T9
T1	84.69	0.00	0.00	11.03	0.00	1.86	0.81	0.00	1.61
T2	9.05	53.49	6.67	22.30	0.00	4.13	0.00	0.00	4.37
T3	0.00	1.75	94.64	1.75	0.00	0.00	0.00	0.00	1.85
T4	0.49	0.00	0.36	93.85	1.38	0.00	0.00	0.00	3.91
T5	0.42	0.69	0.00	0.00	84.47	0.09	0.26	0.00	14.07
T6	0.74	0.37	0.00	0.69	0.23	91.95	0.00	0.00	6.02
T7	0.10	0.00	0.00	0.11	1.37	0.08	82.12	0.00	16.22
T8	0.76	0.14	0.00	0.00	0.41	1.52	0.38	78.73	18.07
T9	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	100.00

Table 14. Localization raw counts (integer frames)

Actual / predicted	T1	T2	T3	T4	T5	T6	T7	T8	T9
T1	614	0	0	24	0	3	3	0	52
T2	10	163	2	231	0	6	0	0	81
T3	0	2	253	2	0	0	0	0	9
T4	6	0	6	1019	13	0	0	0	48
T5	6	8	0	0	1034	1	3	0	411
T6	6	3	0	3	1	882	0	0	26
T7	1	0	0	2	17	1	712	0	291
T8	2	1	0	0	3	4	1	421	271
T9	0	0	0	0	0	0	0	0	496

Taxel 2 has the lowest weighted recall, 53.49%; 22.30% of its weighted predictions are assigned to T4. The next largest systematic routes include T8→T9 (18.07%), T7→T9 (16.22%) and T5→T9 (14.07%). Taxel 9 has 100.00% recall, which does not mean every prediction of T9 is correct: other taxels are also assigned to T9.

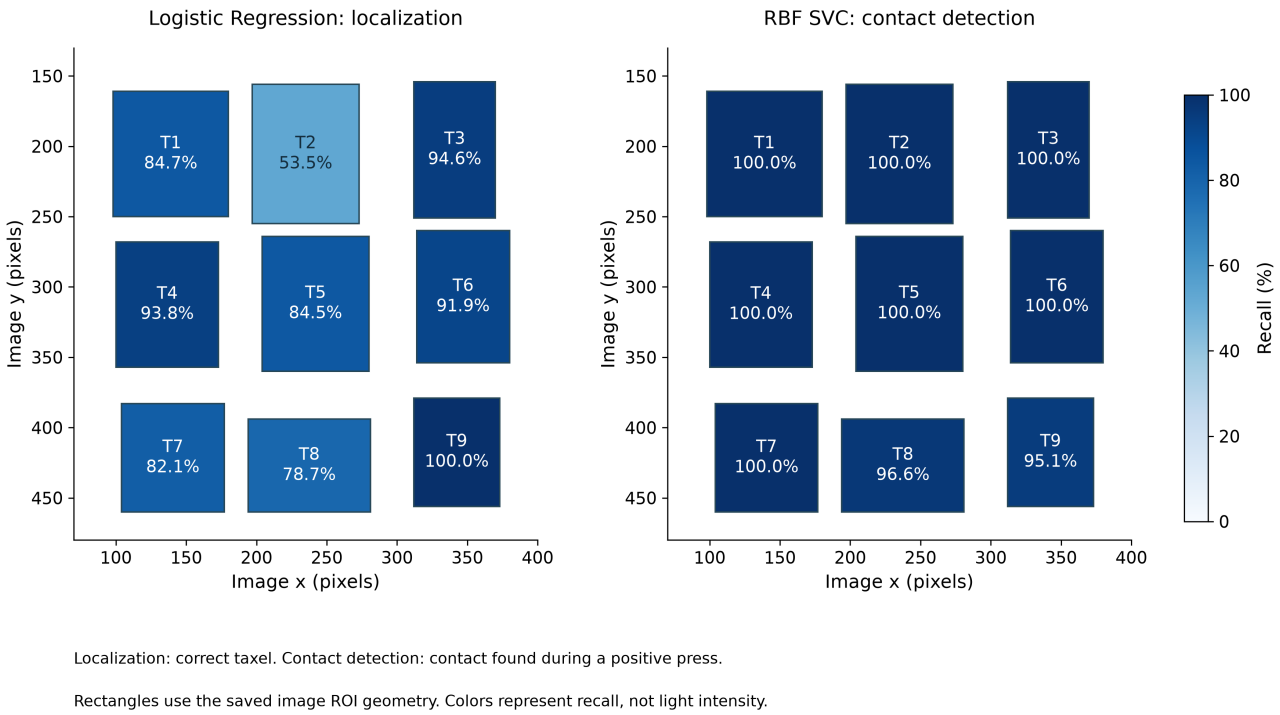


Figure 3. Winning-model recall mapped to the nine physical image ROIs. Left: Logistic-regression localization recall. Right: RBF-SVC positive-contact recall. Each taxel averages six held-out recording recalls. The rectangles retain saved pixel geometry. No-contact specificity is a separate class-level result, not a per-pressed-taxel value.

Table 15. Exact taxel support and recall for the two winning classifiers

Taxel	Eligible frames	Localization recall (%)	Contact recall (%)
1	696	84.69	100.00
2	493	53.49	100.00
3	266	94.64	100.00
4	1092	93.85	100.00
5	1463	84.47	100.00
6	921	91.95	100.00
7	1024	82.12	100.00
8	703	78.73	96.65
9	496	100.00	95.14

8. Contact-detection confusion matrices

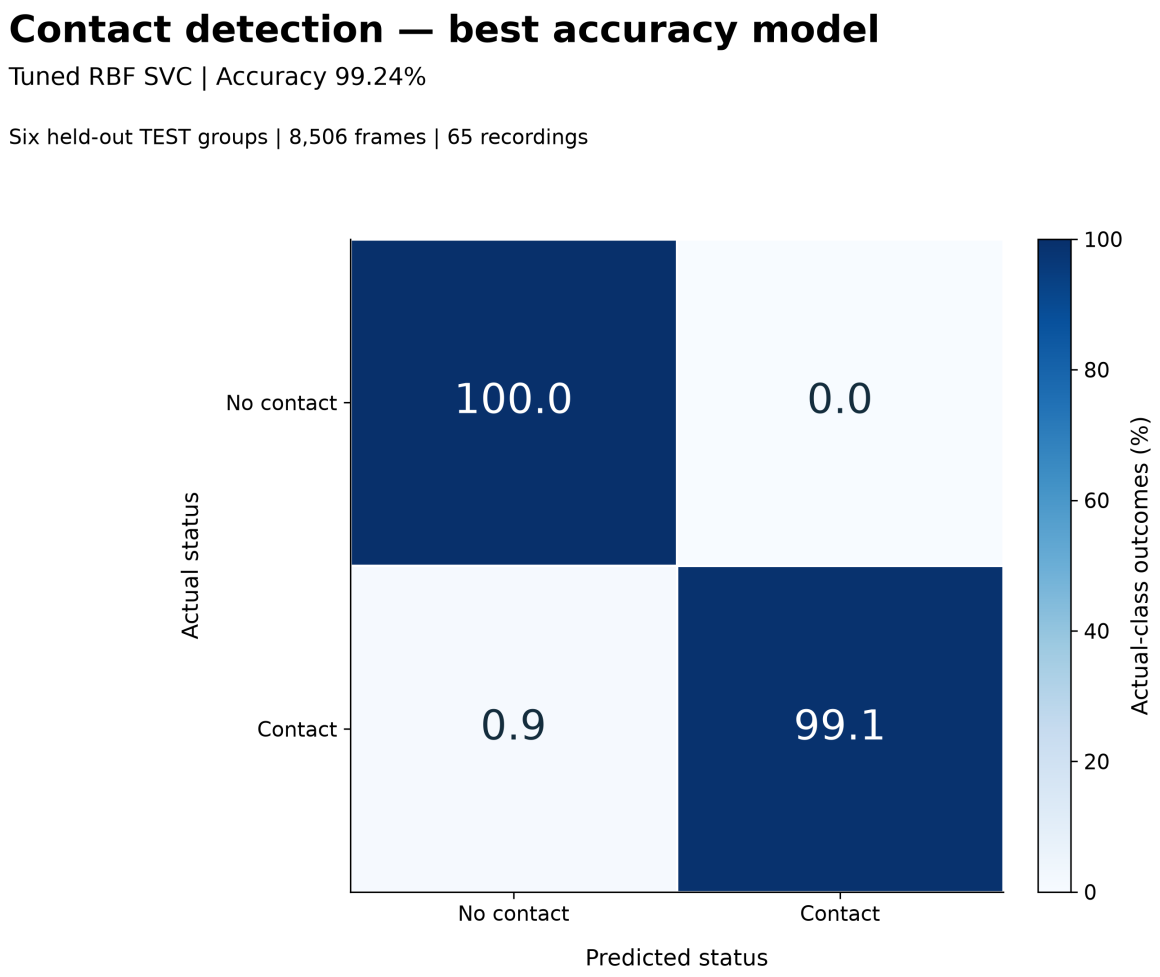
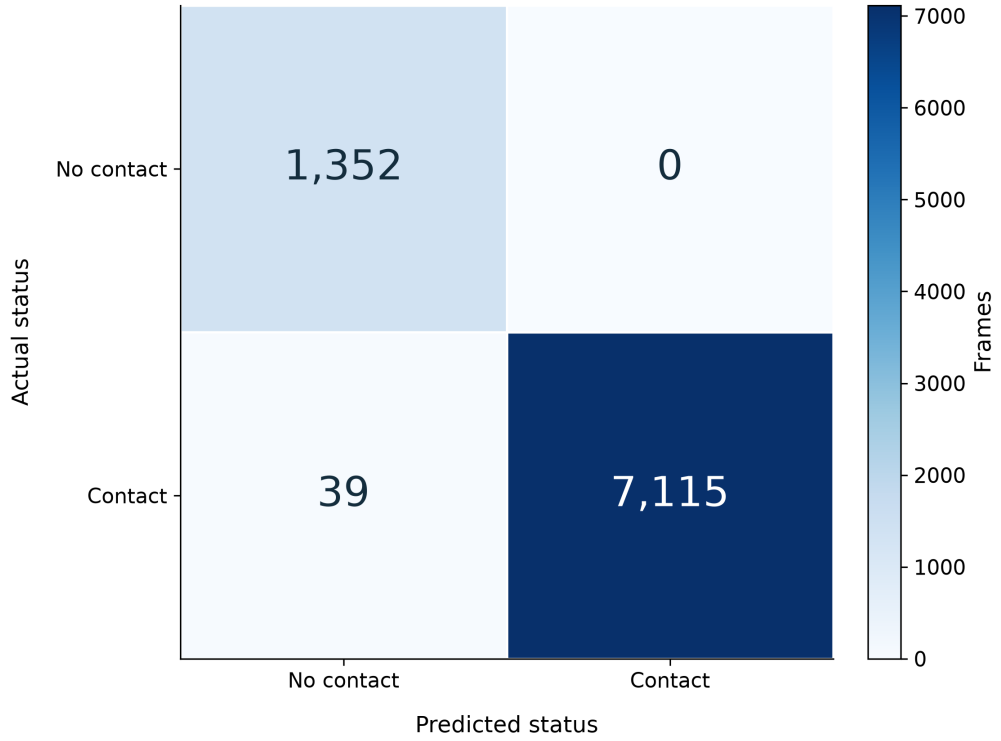


Figure 4. Contact detection: recording-weighted row-normalized 2 × 2 matrix. Tuned RBF SVC. Rows are actual labels and columns predicted labels. No-contact specificity is 100.00%; contact recall is 99.09%. Overall accuracy is 99.24%, while balanced accuracy is 99.54%. Six held-out groups, 8,506 frames and 65 recordings.

Contact detection — best accuracy model

Tuned RBF SVC | Accuracy 99.24%

Six held-out TEST groups | 8,506 frames | 65 recordings



RAW COUNTS — unweighted integer frames

Manual calibration only | Positive force range: 0.05–3 N

Accuracy uses equal recording weights; raw counts separately show every evaluated frame.

Figure 5. Contact detection: separately labeled raw frame counts. The 1,352 dedicated no-contact frames contain 1,352 true negatives and 0 false positives. The 7,154 eligible press frames contain 39 false negatives and 7,115 true positives. These are raw counts; the headline percentages use equal recording weights.

Contact matrix values and full metric set

Table 16. Contact weighted row-normalized percentages (%)

Actual / predicted	No contact	Contact
No contact	100.00	0.00
Contact	0.91	99.09

Table 17. Contact raw counts (integer frames)

Actual / predicted	No contact	Contact
No contact	1352	0
Contact	39	7115

Table 18. All headline contact metrics for Tuned RBF SVC

Metric	Value (%)
Accuracy	99.24
Balanced accuracy	99.54
Contact recall / sensitivity	99.09
Contact precision	100.00
No-contact specificity	100.00
False-positive rate	0.00
Contact-class F1	99.54
Two-class macro F1	98.67

Contact-class F1 (99.54%) and macro F1 (98.67%) are different summaries. Macro F1 also includes the no-contact class, whose precision is reduced by missed positive frames. Reporting either simply as “F1” would be ambiguous.

The positive-contact recall is 100.00% for T1–T7, 96.65% for T8 and 95.14% for T9. Zero observed false positives supports 100.00% specificity on these 11 controls; it does not establish a universal zero false-alarm rate. The dataset does not directly validate contact onset timing, intermittent releases or all excluded low-force/high-force frames.

As a supplementary combined positive-frame measure, contact must be detected and the taxel must be correct on the same held-out frame. Its recording-weighted success is **84.23%** across the 7,154 eligible press frames. This is a joint success metric, not a replacement for either task’s standalone accuracy.

9. Force errors and variation across held-out groups

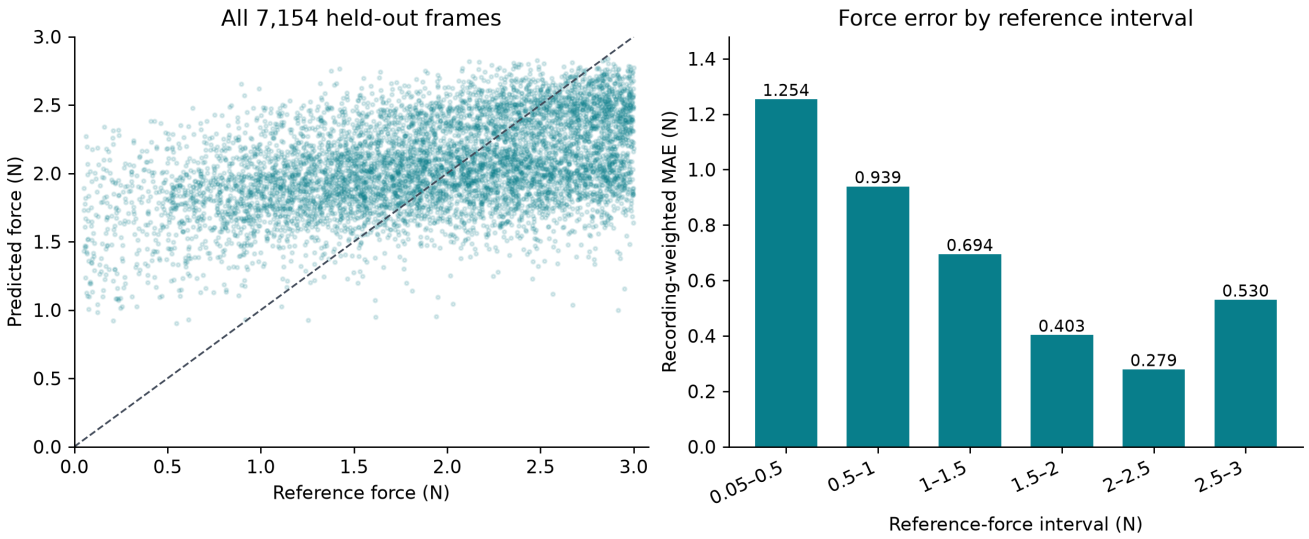


Figure 6. Winning force ensemble: predicted force and error by reference interval. Scatter: all held-out frames, with a dashed identity line; point density is descriptive and unweighted. Bars: recording-weighted MAE, with recording weights recomputed within each bin. The lowest-force intervals show marked overestimation.

Table 19. Force ensemble errors by reference-force interval

Interval (N)	Frames	Recordings	MAE (N)	RMSE (N)	Bias (N)
[0.05, 0.5)	263	52	1.2538	1.2961	1.2538
[0.5, 1)	672	51	0.9390	1.0113	0.9390
[1, 1.5)	1102	52	0.6944	0.7819	0.6841
[1.5, 2)	1490	53	0.4029	0.4789	0.3008
[2, 2.5)	1749	52	0.2789	0.3437	-0.0572
[2.5, 3]	1878	54	0.5298	0.6372	-0.5236

The overall 0.4972 N MAE is not uniform over the interval. MAE is 1.2538 N below 0.5 N, 0.279 N at 2–2.5 N, and 0.530 N at 2.5–3 N. Positive low-force bias and negative high-force bias suggest compression toward intermediate force values. This pattern is a descriptive finding, not a proven causal explanation of the model behavior.

A low pooled force error does not establish an accurate detection limit, calibrated linearity or acceptable error at every force. Dense upper-range coverage, weighting choices, timing, and manual loading trajectories may all affect model comparisons. A controlled matched-force experiment is needed to separate their effects.

Performance in each complete held-out TEST group

Table 20. Winning-model performance by outer TEST group

TEST	Press frames	Contact frames	Force MAE (N)	Loc. acc. (%)	Loc. F1 (%)	Contact acc. (%)	Contact bal. acc. (%)
1	2365	2609	0.5683	48.41	44.11	100.00	100.00
2	1025	1268	0.4937	88.35	87.97	99.29	99.56
3	686	939	0.5056	96.00	95.97	98.68	99.19
4	1117	1361	0.4678	90.86	90.61	98.61	99.15
5	1016	1265	0.4678	98.20	98.21	99.43	99.65
6	945	1064	0.4799	87.46	83.04	99.46	99.70

Localization varies from 48.41% in TEST1 to 98.20% in TEST5. Contact accuracy stays between 98.61% and 100.00%, while force MAE ranges from 0.4678 to 0.5683 N. TEST1 is the major localization weakness despite perfect observed contact detection there. Contact detection and identifying the correct taxel therefore have distinct difficulty profiles.

These six groups are the relevant units for assessing generalization variability. Their training sets overlap, and frames within recordings are correlated. The displayed range is descriptive; it is not a confidence interval. A new independent collection is needed to estimate prospective performance after the final model choice.

Table 21. Force ensemble errors by pressed taxel

Taxel	Frames	MAE (N)	RMSE (N)	Bias (N)	R^2
1	696	0.5960	0.7574	0.0321	0.2160
2	493	0.4333	0.5764	0.0894	0.4636
3	266	0.4708	0.6224	0.1055	0.4310
4	1092	0.5767	0.7003	0.0702	0.0403
5	1463	0.5270	0.6313	0.1395	0.1085
6	921	0.5021	0.6168	-0.0035	0.0573
7	1024	0.4869	0.6076	0.1008	0.1986
8	703	0.4742	0.5881	0.0637	0.3105
9	496	0.4077	0.5420	0.1466	0.5000

All subgroup values on this page were recomputed from the three winning models' saved predictions. They are not copied from the distinct nested-selector procedure. Equal recording weighting is retained within every displayed subgroup.

10. Optical and light-response diagnostics

10.1 Direct HSV V brightness measurements

V is the camera image brightness channel in its 8-bit 0-255 encoding. The direct metrics below quantify the recorded V values themselves: ROI mean brightness, baseline brightness, signed mean-brightness change, temporal variability, observed intensity range and peak pixel value. They are distinct from classifier accuracy and from the positive-only optical response used in the preceding feature analysis. [S16]

The sample is unchanged: 7,154 eligible contact frames from 54 manual press recordings and 1,352 frames from 11 dedicated no-contact recordings. A taxel contact mean averages its six target-press recording means; its no-contact mean averages the 11 control recording means. Each recording has equal weight. These measurements describe the evaluated manual subset, not every archived frame.

Table 22. Direct V-channel metric definitions

Metric	Definition
Raw ROI mean V	m_sjt = mean V over the fixed taxel pixels in one frame; recorded roi_j_mean_v
Baseline mean V	b_sj = mean V across pixels and captured baseline frames, from that recording's baseline_summary.json mean_v
Contact mean V	Average m_sjt over eligible contact frames within each recording, then average six recording means
Signed mean-brightness change	For each recording, mean(m_sjt) - b_sj; then average six recordings. Negative values are retained
Mean positive pixel change	Mean of max(V_pixel - baseline_median_pixel, 0); this existing positive-only statistic is not the signed mean-brightness change
Temporal noise of raw V	RMS of within-control-recording sample SDs of m_sjt, ddof=1; 11 control recordings
No-contact CV	100 x temporal noise SD / no-contact mean V; a relative temporal-stability descriptor

Table 23. Direct brightness and baseline-referenced change, in camera V levels

Taxel	Baseline mean V	No-contact mean V	Contact mean V	Signed change V	Positive pixel change V
1	27.404	26.199	27.017	-0.386	1.029
2	27.388	26.091	26.846	-0.542	0.939
3	27.576	26.419	27.939	0.362	1.564
4	27.795	26.549	27.308	-0.487	0.959
5	27.883	26.504	27.238	-0.645	0.857
6	27.820	26.574	27.535	-0.285	1.135
7	27.782	26.571	27.636	-0.146	1.162
8	27.693	26.576	27.997	0.304	1.499
9	27.716	26.755	28.085	0.370	1.523

Mean raw V spans 26.091-26.755 without contact and 26.846-28.085 during eligible contact. The baseline column refers to the six press recordings' own unloaded captures. Dedicated no-contact recordings are separate acquisitions and are not substituted for those per-recording baselines.

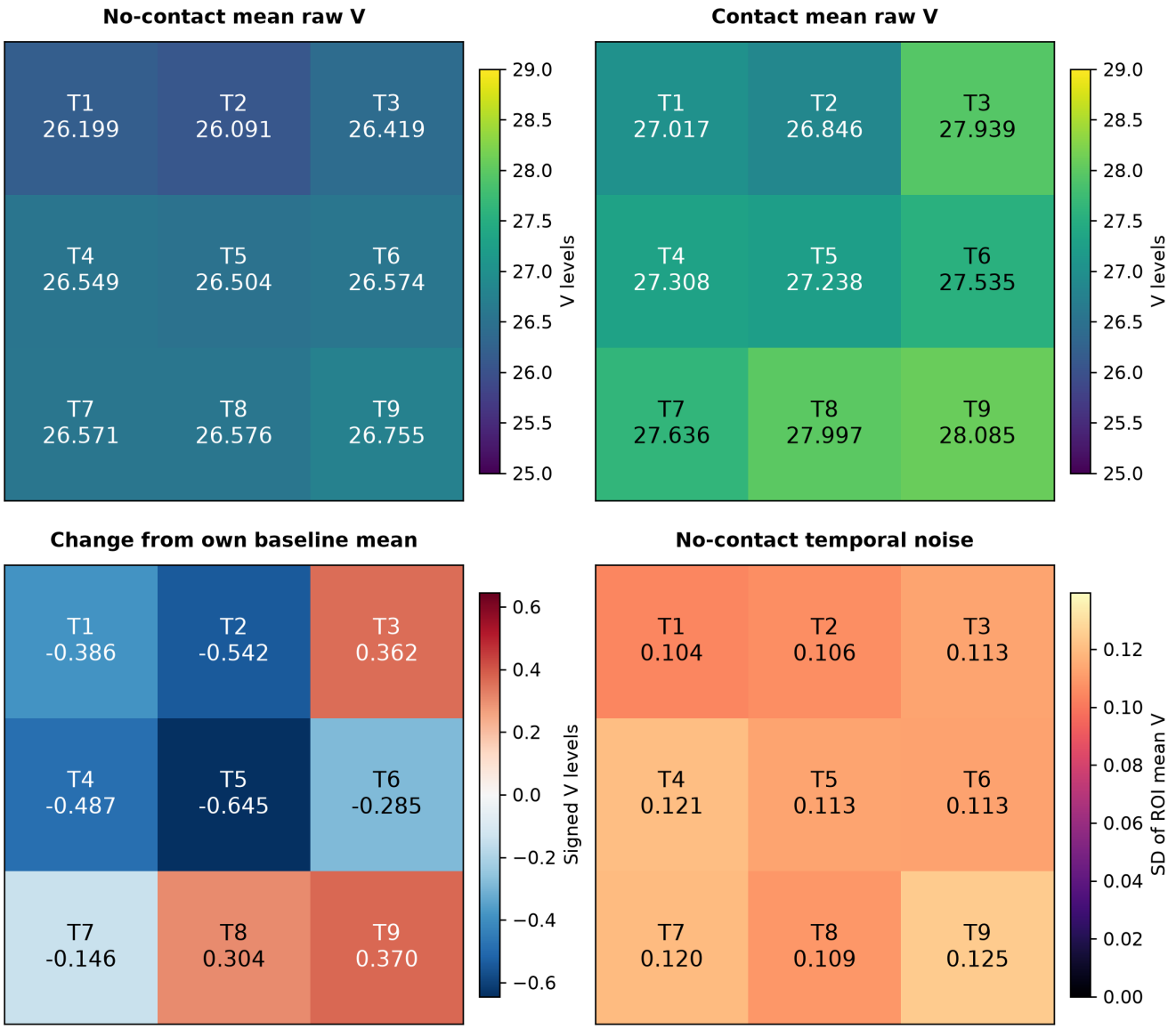


Figure 7. Direct raw-V brightness, signed baseline change and temporal noise. Each 3 x 3 map follows taxel order T1-T9. Raw contact and no-contact panels share the same 25-29 V scale, so their colors are comparable. The signed-change panel retains both decreases and increases from the per-recording baseline-capture mean. Noise is the temporal SD of ROI mean V, not the SD of positive-clipped delta V.

A positive-only response can coexist with a negative mean-brightness change. Some pixels brighten while others darken, and the baseline may differ from the recording's later brightness level. Clipping negative pixel changes discards the decreases; taking an ROI mean first retains them. These two operations answer different questions and must not share one ambiguous "delta V" label.

The baseline mean in the direct-V table is the saved mean over baseline frames and pixels. The positive-change feature uses the saved per-pixel median baseline. They are explicitly different reference constructions. The signs in this report are measured outcomes; a source of drift or illumination change is not established by these descriptive comparisons.

Raw-V variability, ranges and pixel peaks

Table 24. Temporal variation and observed contact brightness range

Taxel	Control noise SD V	Contact temporal SD V	Control CV (%)	Min ROI mean V	Max ROI mean V	Peak pixel V
1	0.1045	0.6140	0.399	25.304	35.904	132
2	0.1064	0.5521	0.408	25.258	33.053	47
3	0.1131	0.8815	0.428	25.600	34.359	72
4	0.1208	0.5452	0.455	25.506	33.952	57
5	0.1132	0.5080	0.427	25.522	33.764	55
6	0.1127	0.7443	0.424	25.598	35.772	163
7	0.1197	0.7102	0.450	25.590	34.404	90
8	0.1094	0.8358	0.411	25.534	34.971	59
9	0.1246	0.8581	0.466	25.738	34.461	46

No-contact temporal noise of raw ROI mean V ranges from 0.1045 to 0.1246 V levels, with CV 0.399-0.466%. This quantifies short-term within-recording variability. It is not a pixel-level spatial standard deviation, a cross-recording SD, or the baseline_summary std_v field, which pools baseline pixels and frames.

Contact temporal SD includes intentional force-related variation and therefore is not labeled sensor noise. Min/max ROI means are extrema over eligible frames; peak pixel V is the largest recorded ROI pixel maximum. A peak of 163 at T6, for example, does not imply an ROI mean of 163. No target peak reaches the encoded maximum of 255 in the evaluated press subset.

Table 25. Additional direct-V summaries

Taxel	Median of recording medians V	Baseline change (%)	Contact-control contrast V	Contact mean /255 (%)
1	25.965	-1.518	0.818	10.595
2	25.915	-2.036	0.755	10.528
3	27.195	1.276	1.520	10.956
4	26.360	-1.818	0.759	10.709
5	26.353	-2.350	0.734	10.682
6	26.577	-1.100	0.961	10.798
7	26.862	-0.566	1.065	10.838
8	27.350	1.105	1.422	10.979
9	27.378	1.337	1.331	11.014

Baseline-change percent averages 100 x (recording mean V - own baseline mean V) / own baseline mean V. The contact-control contrast compares independent recording groups and is not a paired causal contact effect. The /255 percentage describes the digital encoding only; it is not percent optical power, calibrated illuminance or sensor accuracy.

These raw V metrics, together with positive-only response, active area and spatial distribution below, provide a direct brightness characterization. Physical sensitivity in V/N, calibrated lux, long-term drift and a detection limit require separate controlled measurements.

10.2 Positive optical response and spatial distribution

Light diagnostics use the original manual acquisition frame_features.csv files matched exactly to the same 8,506 held-out frame identities. All 65 source CSVs were found; archive lineage, source hashes, unique frame joins and timestamps were checked. A stable identity combines archive hash, recording directory, zero-based row/video index and capture-frame identifier. [S10–S12]

These acquisition-time CSV measurements are a companion optical analysis. Model training used video-recomputed features, so acquisition values are not claimed to be numerically identical to compressed-video measurements. The optical diagnostics are not an additional model training run.

Table 26. Light metrics supported by the available data

Metric	Exact construction and aggregation
Positive mean ΔV: D_jt	Mean of per-pixel max(V−baseline,0) over taxel j
Reference floor: b_j,−g	Median of recording medians of D_jt in no-contact controls outside held-out group g
Excess response: E_jt	max(D_jt − b_j,−g, 0); mean within press recording, then equal mean over six recordings per target taxel
Temporal noise: σ_j	Square root of the mean of within-recording sample variances (ddof=1) of D_jt across 11 no-contact recordings
Robust noise	Per-control 1.4826 × MAD(D_jt); median across control recordings
Response-to-noise proxy	Each press recording's mean target E divided by RMS control SD outside its held-out group; mean over six presses
Active-area fraction	Fraction of ROI pixels with ΔV≥10; average within recording, then over six presses; report as %
Spatial response share	Average E_j over all eligible frames in a recording; normalize nine means to sum to 100%; average six profiles per pressed taxel
Force–light association	Spearman correlation of target D_jt with the existing lag-aligned force within each recording; median over six records per taxel
Drift / clipping diagnostics	No-contact raw-V last-quartile median minus first-quartile median; target raw maximum reaching 255 is a frame flag

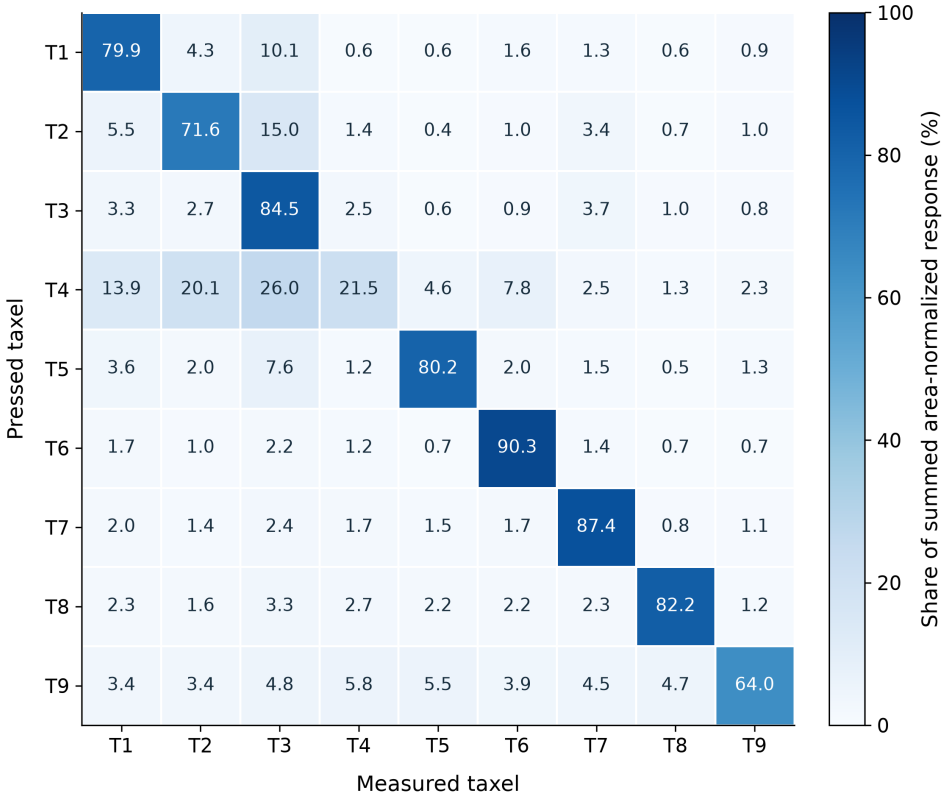
All 7,154 eligible press frames remain in the distribution calculation, including 1,153 with zero total excess. Each recording has a nonzero mean-response denominator. The spatial mean-then-normalize rule avoids dropping zero-response frames or implicitly weighting recordings by their length.

Camera V levels are not lux or optical power. The response-to-noise ratio is an exploratory proxy, not calibrated photometric SNR. Off-target response is a spatial-spread/cross-talk proxy, not a measured mechanical transfer coefficient. Different manual force histories mean taxel averages cannot establish intrinsic sensitivity at matched force.

Optical response distribution across nine taxels

Spatial spread / cross-talk proxy from recorded optical measurements

7,154 press frames | Six recordings per pressed taxel | Reference floors exclude held-out groups



Per recording: average excess mean ΔV for each taxel, then normalize the nine values to 100%.

Each row averages six recording profiles. Values describe response spread, not prediction accuracy.

All 7,154 frames retained, including 1,153 zero-total frames. Camera units, not optical power.

Figure 8. Spatial distribution of excess optical response. Rows identify the pressed taxel; columns identify the measured taxel. Values are recording-averaged shares of the nine mean excess responses, in percent. This is an optical response matrix, not a classifier confusion matrix. Taxel 4 spreads strongly toward T1–T3.

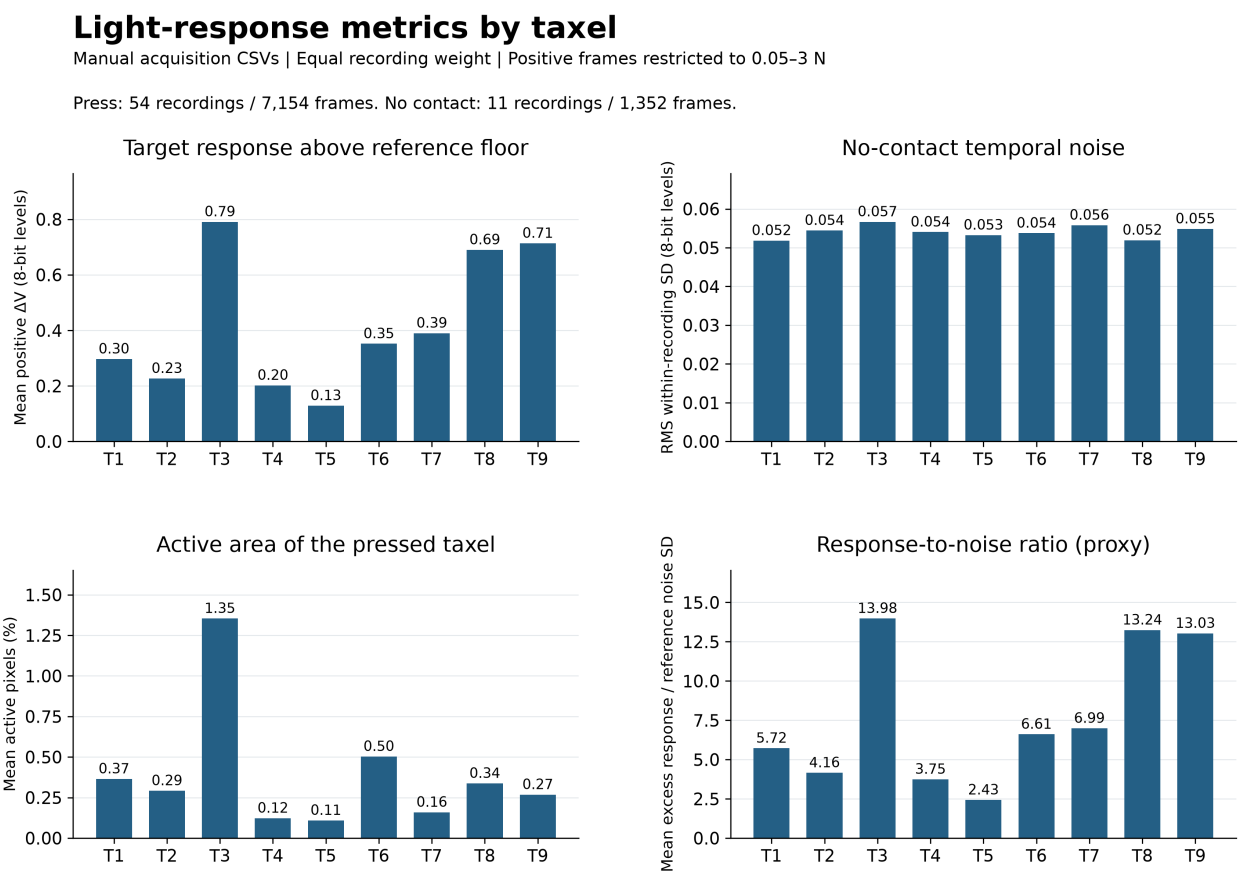


Figure 9. Optical response, temporal noise, active area and response-to-noise proxy. Press measures average six recording means for each target taxel. Descriptive control noise uses all 11 controls; the response/noise proxy uses only controls outside each press group. Units and aggregation are defined in Section 10.

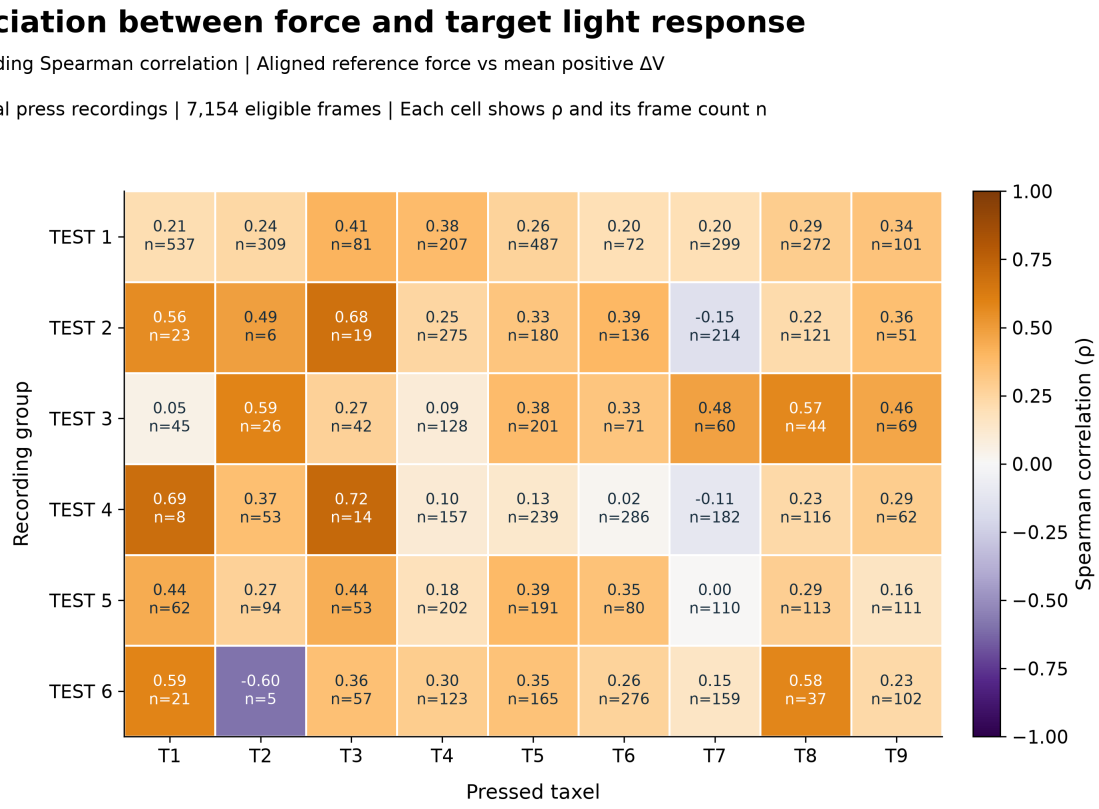


Figure 10. Within-recording force–light Spearman correlations. Each cell compares the target taxel's acquisition-time mean positive ΔV with the pre-existing aligned force in one TEST/taxel recording. No additional lag or signal threshold was optimized. The fixed −1 to +1 scale preserves negative associations; temporal dependence precludes independent-frame significance claims.

Light measurements as searchable tables

Table 27. Target optical measurements by taxel

Taxel	Mean positive ΔV	Mean excess ΔV	Noise SD ΔV	Robust σ ΔV	Active area (%)	Response / noise
1	1.0291	0.2969	0.0518	0.0515	0.3651	5.723
2	0.9392	0.2268	0.0544	0.0568	0.2921	4.158
3	1.5637	0.7910	0.0567	0.0590	1.3542	13.980
4	0.9592	0.2022	0.0540	0.0513	0.1229	3.745
5	0.8574	0.1291	0.0532	0.0488	0.1093	2.428
6	1.1351	0.3519	0.0537	0.0536	0.5033	6.610
7	1.1625	0.3900	0.0558	0.0554	0.1593	6.987
8	1.4990	0.6901	0.0519	0.0498	0.3374	13.244
9	1.5233	0.7137	0.0549	0.0562	0.2682	13.029

Table 28. Response concentration and within-recording correlation summaries

Taxel	Target share (%)	Off-target share (%)	Median ρ	Min ρ	Max ρ
1	79.92	20.08	0.500	0.055	0.690
2	71.57	28.43	0.317	-0.600	0.593
3	84.53	15.47	0.422	0.269	0.719
4	21.50	78.50	0.215	0.095	0.378
5	80.23	19.77	0.339	0.128	0.387
6	90.27	9.73	0.298	0.023	0.395
7	87.42	12.58	0.078	-0.155	0.484
8	82.17	17.83	0.290	0.221	0.584
9	63.98	36.02	0.317	0.163	0.461

No-contact noise spans 0.0518–0.0567 camera V levels. Mean target excess spans 0.1291–0.7910 levels; response/noise proxy spans 2.43–13.98; active area spans 0.1093–1.3542%. These are observed descriptive ranges, not validated detection limits.

Taxel 4 retains only 21.50% of total excess response in its own region, yet has 93.85% localization recall. The classifier can exploit the pattern across all regions; its accuracy need not track target-only optical concentration. Taxel 7 has the lowest median force–light correlation (ρ = 0.078), while T1 has the highest (ρ = 0.500).

Exact optical matrix values

Table 29. Excess-response distribution, mean recording-normalized shares (%)

Actual / predicted	T1	T2	T3	T4	T5	T6	T7	T8	T9
T1	79.92	4.34	10.08	0.63	0.62	1.55	1.34	0.62	0.91
T2	5.52	71.57	14.99	1.41	0.41	0.98	3.41	0.74	0.96
T3	3.26	2.67	84.53	2.50	0.64	0.88	3.72	1.01	0.79
T4	13.95	20.07	26.04	21.50	4.56	7.79	2.48	1.28	2.32
T5	3.63	1.95	7.64	1.24	80.23	1.96	1.54	0.47	1.33
T6	1.73	1.04	2.15	1.23	0.75	90.27	1.43	0.66	0.73
T7	1.99	1.41	2.35	1.72	1.54	1.66	87.42	0.80	1.11
T8	2.34	1.65	3.27	2.67	2.20	2.20	2.26	82.17	1.24
T9	3.40	3.42	4.84	5.80	5.53	3.87	4.47	4.69	63.98

Table 30. Spearman ρ for every TEST group and taxel

TEST	T1	T2	T3	T4	T5	T6	T7	T8	T9
1	0.210	0.239	0.406	0.378	0.264	0.196	0.196	0.294	0.340
2	0.557	0.486	0.675	0.246	0.333	0.395	-0.155	0.221	0.363
3	0.055	0.593	0.269	0.095	0.380	0.332	0.484	0.573	0.461
4	0.690	0.366	0.719	0.103	0.128	0.023	-0.108	0.230	0.294
5	0.442	0.268	0.437	0.183	0.387	0.346	0.003	0.285	0.163
6	0.590	-0.600	0.355	0.296	0.346	0.264	0.153	0.584	0.232

The correlation calculation has 5–537 paired eligible frames per recording, with different temporal dependence and force trajectories. Small-sample cells can be unstable. No p-values, causal sensitivity or linearity claims are inferred from these correlations.

The recorded target maximum-V flag never reaches 255 in the evaluated press subset. This indicates no observed target-ROI raw maximum of 255 in these CSV rows; it does not prove that the complete optical system is free of saturation. Median no-contact raw-V quartile drift ranges from −0.0272 to +0.0126 levels across taxels. Neither value supplies long-term stability or recalibration intervals.

11. Exact winning model configurations

The outer-fold configurations below generated the headline scores. Configuration IDs refer to the exhaustive grid in Appendix A. All members use weighted StandardScaler fitted on the corresponding training rows. Lag/floor fitting is training-only. The output-history index h maps 0→0 s, 1→0.3 s, 2→1.0 s. [S1, S8]

Table 31. Winning classifier configuration in each held-out fold

TEST	Localization ID / pack	Loc. τ (s)	Contact ID / pack	Contact τ (s)	Contact threshold
1	loc_048 / temporal	0.3	con_022 / shape	1.0	0.33409756422042847
2	loc_036 / invariant	1.0	con_022 / shape	1.0	0.3497229218482971
3	loc_036 / invariant	1.0	con_022 / shape	1.0	0.32444050908088684
4	loc_036 / invariant	1.0	con_022 / shape	1.0	0.29617786407470703
5	loc_036 / invariant	1.0	con_022 / shape	1.0	0.3024323284626007
6	loc_036 / invariant	1.0	con_022 / shape	1.0	0.3111855387687683

Localization uses LogisticRegression(C=0.03, max_iter=2500, random_state=1741). TEST1 uses temporal features (288 dimensions) and τ=0.3 s. TEST2–TEST6 use invariant features (117 dimensions); their exact output time constants are listed above. Classification uses argmax of smoothed class scores.

Contact uses SVC(C=1, kernel="rbf", gamma=1/90=0.011111111111111112, cache_size=256, decision_function_shape="ovr", random_state=1741) on shape features (90 dimensions). Its score is sigmoid(decision_function), followed by the listed EMA, then contact if score ≥ fold threshold. Thresholds were selected by the balanced-accuracy policy on inner OOF predictions.

Force ensemble composition

Each outer prediction averages three separately tuned force families after their selected output EMA and residual correction. All winning force members use temporal features (288 dimensions). Most folds use for_050, for_056 and for_059; TEST2 uses for_055 in place of for_056. The exact folds, offsets and time constants follow on the next page.

Table 32. Force member estimator settings (additional defaults in Appendix B)

ID	Estimator	Explicit settings
for_050	ExtraTreesRegressor	180 trees; min_samples_leaf=3; max_features=0.75; n_jobs=1; random_state=1741
for_055	HistGradientBoostingRegressor	max_iter=180; learning_rate=0.05; min_samples_leaf=20; max_leaf_nodes=15; loss=squared_error; l2_regularization=10; early_stopping=False; random_state=1741
for_056	HistGradientBoostingRegressor	As for_055, but max_leaf_nodes=31
for_059	XGBRegressor	220 trees; learning_rate=0.04; min_child_weight=5; max_depth=5; reg_lambda=20; tree_method=hist; n_jobs=1; random_state=1741

Exact force corrections and output history

Table 33. All 18 force-ensemble member settings used for held-out scoring

TEST	Candidate	Family	τ (s)	Correction	Offset (N)
1	for_050	Extra Trees	0.0	median	0.11000364382382512
1	for_056	HistGB	0.0	median	0.07882320945789201
1	for_059	XGBoost	0.0	median	0.09625087054912093
2	for_055	HistGB	0.0	median	0.10901741162705457
2	for_050	Extra Trees	0.0	median	0.10593629488925335
2	for_059	XGBoost	0.0	median	0.0970969430474912
3	for_056	HistGB	0.0	median	0.09374066103661953
3	for_059	XGBoost	0.0	median	0.11504363333800048
3	for_050	Extra Trees	0.0	median	0.14190227448836593
4	for_056	HistGB	0.3	median	0.10363438450784854
4	for_059	XGBoost	0.3	median	0.12177224632432046
4	for_050	Extra Trees	0.0	median	0.1408816589434914
5	for_056	HistGB	0.3	median	0.10187709111648936
5	for_050	Extra Trees	0.0	median	0.11651780258932955
5	for_059	XGBoost	0.0	median	0.0732525453464563
6	for_059	XGBoost	0.3	median	0.11652466294880082
6	for_056	HistGB	0.3	median	0.10905367485922174
6	for_050	Extra Trees	0.0	median	0.16707105034631944

Offsets are added after the selected EMA; the result is clipped at zero. “Median” means the weighted 0.5 quantile of inner OOF residuals y−prediction. The implementation sorts residuals and interpolates at q·Σw against cumulative weight minus half the current weight. It is not an outer-test residual correction.

Training no-contact floors and scaler means/scales vary across folds and are learned preprocessing parameters rather than selected hyperparameters. Their estimator boundary is fixed by whole-recording group assignment; no held-out recording contributes. No single all-data configuration should be substituted for the six fold configurations when describing the source of these performance figures.

Full-pool research configurations available for inference

After evaluation, full-pool research models were fitted. The available Logistic-regression component matches the winning family, while the saved localization manifest still routes the competing ensemble. This report does not change that manifest or the live GUI. These full-data settings are provided for reproducibility; their internal selection scores are not additional held-out test results.

Table 34. Available full-pool best-family components

File / candidate	Pack	τ (s)	Threshold or force offset
force_for_056.joblib	temporal	0.3	offset=0.10292978395171193
force_for_050.joblib	temporal	0.0	offset=0.12944881296122535
force_for_059.joblib	temporal	0.3	offset=0.10841652905616302
localization_loc_036.joblib	invariant	1.0	argmax
contact_con_022.joblib	shape	1.0	threshold=0.32685983180999756

Full-pool lag: 300 ms. Positive-light sum floors, in T1–T9 order: [3786, 4026, 2952, 3509, 3999, 3470, 3096, 3218, 2608]. Divide floor-corrected sums by the exact ROI areas before feature construction. Each saved estimator includes its fitted scaler; learned scalers must not be refitted on test data.

Inference order: obtain correctly oriented V measurements and the saved baseline; compute ROI measurements; subtract the stored floor in the specified feature formula; update causal feature histories; apply the stored scaler; generate estimator scores; apply member-specific output history and force correction/contact threshold; average force member outputs; return force, contact and taxel. Reset state between recordings or at the specified time discontinuities.

Exact scalar estimator parameters and relevant saved component hashes are printed in Appendix B. Model binary files retain learned weights/trees and scaler arrays. This document provides the complete scientific description and configuration; it does not replace the trained estimator binaries for running inference.

12. Findings, limitations and next experiments

Findings supported by the measurements

Expanded optical patterns, causal history, hyperparameter tuning and output policies improve the pooled best results over the fixed benchmark. The most accurate localization family is regularized Logistic regression, the most accurate contact family is RBF SVC, and the lowest-MAE force model is a three-family ensemble. Increased model complexity alone is not an explanation: tree and MLP classifiers do not lead localization accuracy.

The improvement is measured for the combined development strategy. The run does not contain a full factorial ablation that isolates the causal contribution of each feature, correction, weighting choice or model. Differences between the frame-only selector and the best family suggest useful temporal information, but are not a controlled estimate of a single EMA effect.

Contact detection is consistently strong within the defined press/control comparison. Localization errors concentrate in TEST1 and specific taxel confusions, especially T2. Force regression remains the weakest task: overall R^2 is 0.3113 and low-force errors are large. Optical distributions show measurable off-target response, but classifiers can use distributed patterns successfully.

Limits that must remain in the thesis

The dataset contains one archived manual collection with six TEST groups and 11 no-contact controls, not an independent multi-device or multi-day prospective trial. Family ranking uses observed outer results, so winner selection can be optimistic. Temporal smoothing benefits stationary recorded contacts; moving and multiple simultaneous contacts were not evaluated. Contact prevalence, dedicated-control design and restricted force eligibility limit transfer to continuous deployment.

Reference-force uncertainty, physical taxel coordinates in millimetres, calibrated illumination, controlled matched-force sweeps, onset/release annotations and deployment latency were not established in this analysis. Accordingly, this report does not claim millimetre localization error, lux response, force detection limit, hysteresis, full-scale linearity, dynamic response time or a validated operating range.

Specific next experiments

Freeze the three headline model families and all preprocessing before collecting a new manual test dataset. Repeat complete sessions across days, camera settings and mechanical reassemblies; include continuous no-contact periods, moving contacts and controlled onset/release transitions. Keep new test recordings outside all threshold and hyperparameter selection.

For force, collect balanced repeated loading/unloading plateaus near 0.05–1 N and throughout 1–3 N; preserve loading direction and synchronized timing. For localization, prioritize T2 and the TEST1 acquisition conditions and record matched-force contacts to separate lighting spread from force effects. For contact, evaluate false alarms per unit time and missed events once episode annotations exist. Benchmark the exact intended winner stack with camera acquisition and feature extraction included before making latency claims.

13. Verification and computational reproducibility

Training artifacts record a successful validation audit covering 114 aggregate metric comparisons (maximum absolute discrepancy approximately 1.25×10^{-15}), 30 inner boundaries, source lineage, whole-recording splits, finite scalers, model checksums, refitted fold-1 predictions, label-independent inference, causal-prefix and future-invariance tests, state reset, and batch/streaming agreement. [S9]

The optical analysis records 430 source validation checks and 120 analysis checks, with additional response-distribution verification. For this consolidation, the best-model prediction rows were loaded independently, all headline metrics recomputed, and both raw and weighted normalized matrices reproduced. New fold, force-bin and force-taxel diagnostics use those same winner rows. Existing artifacts were hashed before report generation and checked for preservation. [S1, S10–S13]

Table 35. Recorded experiment environment

Component	Version / setting
Python	3.12.14; Windows 11 build 26200; 24 logical CPUs
NumPy / pandas	2.5.1 / 3.0.5
scikit-learn / XGBoost	1.9.0 / 3.4.0
SciPy / joblib	1.18.0 / 1.5.3
threadpoolctl / PyArrow	3.6.0 / 21.0.0
pytest	9.1.1
Random seeds	1729 fixed benchmark; 1741 tuning
Tuned estimator execution	n_jobs=1 where explicitly set; threadpool_limits(limits=1) during fit

A stored timing audit measured the original saved multi-task bundle, including the competing localization ensemble and alternate contact output: streaming median 41.10 ms and p95 44.46 ms, excluding camera acquisition and pixel feature extraction. That is not a new measurement of the standalone winning-family stack; it must not be cited as its deployed end-to-end latency. [S14]

The immutable feature partitions were not re-read during this report update because their Windows access permissions deny access in the current session. The report relies on preserved training provenance, prediction tables, executed source snapshots and exact configuration exports; the raw archive was accessible in the optical audit. This update consolidates and verifies saved evidence without retraining or changing deployment.

Reproduction sequence

1. Verify the manual archive hash and exclude automatic/replay-only lineage. 2. Extract the original-video ROI feature contract and saved baselines. 3. Preserve whole-recording TEST assignments and reproduce fold-local force alignment. 4. Fit floor/scaler and candidates only on each training boundary. 5. Apply inner selection and score each outer group once. 6. Pool predictions with recording weights and reproduce the comparison tables. 7. Rank families by the report's stated criterion. 8. Produce confusion and optical figures from the corresponding saved rows. 9. Use independent new recordings for prospective validation.

14. Ready-to-use ChatGPT Writing instructions

The following prompt can be pasted into ChatGPT Writing together with this PDF. It specifies the evidence hierarchy and prevents accidental mixing of different model scores.

Writing prompt

Use the attached “Manual Calibration: Complete Technical Report” as the technical source for my thesis. Follow a standard thesis structure: Introduction and objectives, Related work, System and experimental design, Methodology, Results, Discussion, Conclusions and future work. Write formal academic prose with clear equations and numbered tables/figures. Keep methodological detail sufficient for replication.

Use the following models consistently in the abstract, results, discussion and conclusion: localization = Tuned Logistic regression, 84.88% accuracy and 85.00% macro F1; contact detection = Tuned RBF SVC, 99.24% accuracy, 99.54% balanced accuracy, 99.09% recall, 100.00% precision and specificity, 99.54% contact-class F1 and 98.67% macro F1; force = Tuned ensemble, MAE 0.4972 N, RMSE 0.6299 N, bias +0.0827 N and R^2 0.3113. Do not combine metrics from different models or substitute internal selection scores for held-out scores.

State that only Manual Calibration (2).zip was used. Evaluation used six complete TEST groups, 7,154 eligible press frames from 54 recordings for force/localization and 8,506 frames from 65 recordings for contact. Explain equal recording weighting, the inclusive 0.05–3 N interval, train-only preprocessing, nested group selection, causal history and the retrospective final family ranking. Keep all competing models in comparison tables while emphasizing the three headline winners.

Include the 9 × 9 localization and 2 × 2 contact matrices in separately labeled row-normalized and raw-count views. Rows are actual labels; columns predicted labels. The normalized matrices use recording weights, so their values cannot be reconstructed by dividing raw counts. Include ROI performance heatmaps, the optical response-distribution matrix and light diagnostics. Distinguish optical response share from classification accuracy, and camera V units from lux or calibrated optical power.

Use the exact ROI coordinates, segmentation threshold, feature definitions, fold configurations, training seeds, candidate grid and estimator parameters in this report. Describe the evaluation fold configurations separately from the available full-data research fit. Do not say the winning models were deployed in the live GUI. Force is regression; do not invent a force-accuracy percentage.

Report direct HSV V brightness metrics from Section 10.1: raw mean V, own baseline-capture mean V, signed mean-brightness change, raw-V temporal noise and CV, ROI-mean ranges and peak pixel V. Keep these separate from the mean positive-clipped per-pixel change and excess-response metrics in Section 10.2. Preserve negative baseline changes and distinguish temporal SD from spatial pixel SD. Do not reinterpret digital V levels or /255 percentages as lux, optical power or model accuracy.

Discuss TEST1 localization weakness, T2 confusion, low-force overestimation, limited force R^2, dedicated control recordings, temporal dependence, the restricted force interval and the lack of untouched prospective validation. Attribute improvements to the combined search strategy; do not invent causal ablation results, statistical significance, confidence intervals, physical detection limits, calibrated sensitivity, millimetre errors or end-to-end latency.

Use this report’s project source IDs as internal evidence. For the Related work chapter, obtain and verify scholarly sources separately; do not invent authors, publications, hardware specifications, citations or unrecorded experimental details. Mark genuinely missing thesis information for the researcher to supply. Preserve every supplied numeric result and its units. End with specific future experiments grounded in the reported limitations.

Appendix A. Exhaustive tuning configuration grid

This appendix contains all 178 estimator/feature candidates. Each row specifies its task-local ID, family, feature pack, training-weight code and varied parameters. Common constructor settings in Appendix B and feature definitions in Section 3 complete the configuration. Grid membership does not imply that every candidate has its own reported outer test row; inner selection produces the 50 family/procedure comparisons.

Weight code “session” means recording weights, with additional class balancing for contact. “bins” is force bin/recording balancing. Feature dimensions: positive=9, optical=27, shape=90, spatial=180, invariant=117, temporal=288. Parameters such as gamma_multiplier are translated by the factory: SVC gamma = multiplier / feature dimension.

Force — 62 candidates

Table 36. Complete force candidate grid

ID	Family	Features	Weights	Varied parameters
for_000	Extra Trees	positive	bins	min_samples_leaf=10; max_features=1.0
for_001	Extra Trees	positive	session	min_samples_leaf=10; max_features=1.0
for_002	Extra Trees	optical	session	min_samples_leaf=3; max_features=0.75
for_003	Extra Trees	optical	session	min_samples_leaf=15; max_features=0.75
for_004	Extra Trees	optical	session	min_samples_leaf=30; max_features=1.0
for_005	Random Forest	optical	session	min_samples_leaf=5; max_features=0.75
for_006	Random Forest	optical	session	min_samples_leaf=20; max_features=0.75
for_007	HistGB	optical	session	max_leaf_nodes=15; loss="squared_error"; l2_regularization=10
for_008	HistGB	optical	session	max_leaf_nodes=31; loss="squared_error"; l2_regularization=10
for_009	HistGB	optical	session	max_leaf_nodes=15; loss="absolute_error"; l2_regularization=10
for_010	XGBoost	optical	session	max_depth=3; reg_lambda=20
for_011	XGBoost	optical	session	max_depth=5; reg_lambda=20
for_012	RBF SVR	optical	session	C=2; epsilon=0.05; gamma="scale"
for_013	RBF SVR	optical	session	C=20; epsilon=0.05; gamma="scale"
for_014	Extra Trees	shape	session	min_samples_leaf=3; max_features=0.75
for_015	Extra Trees	shape	session	min_samples_leaf=15; max_features=0.75
for_016	Extra Trees	shape	session	min_samples_leaf=30; max_features=1.0
for_017	Random Forest	shape	session	min_samples_leaf=5; max_features=0.75
for_018	Random Forest	shape	session	min_samples_leaf=20; max_features=0.75
for_019	HistGB	shape	session	max_leaf_nodes=15; loss="squared_error"; l2_regularization=10
for_020	HistGB	shape	session	max_leaf_nodes=31; loss="squared_error"; l2_regularization=10
for_021	HistGB	shape	session	max_leaf_nodes=15; loss="absolute_error"; l2_regularization=10
for_022	XGBoost	shape	session	max_depth=3; reg_lambda=20
for_023	XGBoost	shape	session	max_depth=5; reg_lambda=20
for_024	RBF SVR	shape	session	C=2; epsilon=0.05; gamma="scale"
for_025	RBF SVR	shape	session	C=20; epsilon=0.05; gamma="scale"
for_026	Extra Trees	spatial	session	min_samples_leaf=3; max_features=0.75
for_027	Extra Trees	spatial	session	min_samples_leaf=15; max_features=0.75

ID	Family	Features	Weights	Varied parameters
for_028	Extra Trees	spatial	session	min_samples_leaf=30; max_features=1.0
for_029	Random Forest	spatial	session	min_samples_leaf=5; max_features=0.75
for_030	Random Forest	spatial	session	min_samples_leaf=20; max_features=0.75
for_031	HistGB	spatial	session	max_leaf_nodes=15; loss="squared_error"; l2_regularization=10
for_032	HistGB	spatial	session	max_leaf_nodes=31; loss="squared_error"; l2_regularization=10
for_033	HistGB	spatial	session	max_leaf_nodes=15; loss="absolute_error"; l2_regularization=10
for_034	XGBoost	spatial	session	max_depth=3; reg_lambda=20
for_035	XGBoost	spatial	session	max_depth=5; reg_lambda=20
for_036	RBF SVR	spatial	session	C=2; epsilon=0.05; gamma="scale"
for_037	RBF SVR	spatial	session	C=20; epsilon=0.05; gamma="scale"
for_038	Extra Trees	invariant	session	min_samples_leaf=3; max_features=0.75
for_039	Extra Trees	invariant	session	min_samples_leaf=15; max_features=0.75
for_040	Extra Trees	invariant	session	min_samples_leaf=30; max_features=1.0
for_041	Random Forest	invariant	session	min_samples_leaf=5; max_features=0.75
for_042	Random Forest	invariant	session	min_samples_leaf=20; max_features=0.75
for_043	HistGB	invariant	session	max_leaf_nodes=15; loss="squared_error"; l2_regularization=10
for_044	HistGB	invariant	session	max_leaf_nodes=31; loss="squared_error"; l2_regularization=10
for_045	HistGB	invariant	session	max_leaf_nodes=15; loss="absolute_error"; l2_regularization=10
for_046	XGBoost	invariant	session	max_depth=3; reg_lambda=20
for_047	XGBoost	invariant	session	max_depth=5; reg_lambda=20
for_048	RBF SVR	invariant	session	C=2; epsilon=0.05; gamma="scale"
for_049	RBF SVR	invariant	session	C=20; epsilon=0.05; gamma="scale"
for_050	Extra Trees	temporal	session	min_samples_leaf=3; max_features=0.75
for_051	Extra Trees	temporal	session	min_samples_leaf=15; max_features=0.75
for_052	Extra Trees	temporal	session	min_samples_leaf=30; max_features=1.0
for_053	Random Forest	temporal	session	min_samples_leaf=5; max_features=0.75
for_054	Random Forest	temporal	session	min_samples_leaf=20; max_features=0.75
for_055	HistGB	temporal	session	max_leaf_nodes=15; loss="squared_error"; l2_regularization=10
for_056	HistGB	temporal	session	max_leaf_nodes=31; loss="squared_error"; l2_regularization=10
for_057	HistGB	temporal	session	max_leaf_nodes=15; loss="absolute_error"; l2_regularization=10
for_058	XGBoost	temporal	session	max_depth=3; reg_lambda=20
for_059	XGBoost	temporal	session	max_depth=5; reg_lambda=20
for_060	MLP	optical	session	hidden_layer_sizes=[64, 32]; alpha=1.0
for_061	MLP	spatial	session	hidden_layer_sizes=[64, 32]; alpha=1.0

Localization — 64 candidates

Table 37. Complete localization candidate grid

ID	Family	Features	Weights	Varied parameters
loc_000	Logistic regression	optical	session	C=0.03
loc_001	Logistic regression	optical	session	C=0.3
loc_002	Logistic regression	optical	session	C=3
loc_003	Logistic regression	optical	session	C=30
loc_004	RBF SVC	optical	session	C=1; gamma_multiplier=0.1
loc_005	RBF SVC	optical	session	C=1; gamma_multiplier=1.0
loc_006	RBF SVC	optical	session	C=10; gamma_multiplier=0.1
loc_007	RBF SVC	optical	session	C=10; gamma_multiplier=1.0
loc_008	RBF SVC	optical	session	C=100; gamma_multiplier=0.1
loc_009	RBF SVC	optical	session	C=100; gamma_multiplier=1.0
loc_010	Extra Trees	optical	session	min_samples_leaf=2; max_features=0.75
loc_011	Extra Trees	optical	session	min_samples_leaf=10; max_features=0.75
loc_012	Logistic regression	shape	session	C=0.03
loc_013	Logistic regression	shape	session	C=0.3
loc_014	Logistic regression	shape	session	C=3
loc_015	Logistic regression	shape	session	C=30
loc_016	RBF SVC	shape	session	C=1; gamma_multiplier=0.1
loc_017	RBF SVC	shape	session	C=1; gamma_multiplier=1.0
loc_018	RBF SVC	shape	session	C=10; gamma_multiplier=0.1
loc_019	RBF SVC	shape	session	C=10; gamma_multiplier=1.0
loc_020	RBF SVC	shape	session	C=100; gamma_multiplier=0.1
loc_021	RBF SVC	shape	session	C=100; gamma_multiplier=1.0
loc_022	Extra Trees	shape	session	min_samples_leaf=2; max_features=0.75
loc_023	Extra Trees	shape	session	min_samples_leaf=10; max_features=0.75
loc_024	Logistic regression	spatial	session	C=0.03
loc_025	Logistic regression	spatial	session	C=0.3
loc_026	Logistic regression	spatial	session	C=3
loc_027	Logistic regression	spatial	session	C=30
loc_028	RBF SVC	spatial	session	C=1; gamma_multiplier=0.1
loc_029	RBF SVC	spatial	session	C=1; gamma_multiplier=1.0
loc_030	RBF SVC	spatial	session	C=10; gamma_multiplier=0.1
loc_031	RBF SVC	spatial	session	C=10; gamma_multiplier=1.0
loc_032	RBF SVC	spatial	session	C=100; gamma_multiplier=0.1
loc_033	RBF SVC	spatial	session	C=100; gamma_multiplier=1.0
loc_034	Extra Trees	spatial	session	min_samples_leaf=2; max_features=0.75
loc_035	Extra Trees	spatial	session	min_samples_leaf=10; max_features=0.75
loc_036	Logistic regression	invariant	session	C=0.03
loc_037	Logistic regression	invariant	session	C=0.3

ID	Family	Features	Weights	Varied parameters
loc_038	Logistic regression	invariant	session	C=3
loc_039	Logistic regression	invariant	session	C=30
loc_040	RBF SVC	invariant	session	C=1; gamma_multiplier=0.1
loc_041	RBF SVC	invariant	session	C=1; gamma_multiplier=1.0
loc_042	RBF SVC	invariant	session	C=10; gamma_multiplier=0.1
loc_043	RBF SVC	invariant	session	C=10; gamma_multiplier=1.0
loc_044	RBF SVC	invariant	session	C=100; gamma_multiplier=0.1
loc_045	RBF SVC	invariant	session	C=100; gamma_multiplier=1.0
loc_046	Extra Trees	invariant	session	min_samples_leaf=2; max_features=0.75
loc_047	Extra Trees	invariant	session	min_samples_leaf=10; max_features=0.75
loc_048	Logistic regression	temporal	session	C=0.03
loc_049	Logistic regression	temporal	session	C=0.3
loc_050	Logistic regression	temporal	session	C=3
loc_051	Logistic regression	temporal	session	C=30
loc_052	RBF SVC	temporal	session	C=1; gamma_multiplier=0.1
loc_053	RBF SVC	temporal	session	C=1; gamma_multiplier=1.0
loc_054	RBF SVC	temporal	session	C=10; gamma_multiplier=0.1
loc_055	RBF SVC	temporal	session	C=10; gamma_multiplier=1.0
loc_056	RBF SVC	temporal	session	C=100; gamma_multiplier=0.1
loc_057	RBF SVC	temporal	session	C=100; gamma_multiplier=1.0
loc_058	Extra Trees	temporal	session	min_samples_leaf=2; max_features=0.75
loc_059	Extra Trees	temporal	session	min_samples_leaf=10; max_features=0.75
loc_060	HistGB	spatial	session	max_leaf_nodes=15; l2_regularization=10
loc_061	MLP	spatial	session	hidden_layer_sizes=[64, 32]; alpha=1.0
loc_062	HistGB	invariant	session	max_leaf_nodes=15; l2_regularization=10
loc_063	MLP	invariant	session	hidden_layer_sizes=[64, 32]; alpha=1.0

Contact — 52 candidates

Table 38. Complete contact candidate grid

ID	Family	Features	Weights	Varied parameters
con_000	Random Forest	optical	session	min_samples_leaf=2; max_features=0.75
con_001	Random Forest	optical	session	min_samples_leaf=10; max_features=0.75
con_002	Random Forest	optical	session	min_samples_leaf=30; max_features=0.75
con_003	Extra Trees	optical	session	min_samples_leaf=3; max_features=0.75
con_004	Extra Trees	optical	session	min_samples_leaf=15; max_features=0.75
con_005	HistGB	optical	session	max_leaf_nodes=7; l2_regularization=10
con_006	HistGB	optical	session	max_leaf_nodes=15; l2_regularization=10
con_007	Logistic regression	optical	session	C=0.1
con_008	Logistic regression	optical	session	C=10
con_009	RBF SVC	optical	session	C=1; gamma_multiplier=1.0
con_010	RBF SVC	optical	session	C=10; gamma_multiplier=1.0
con_011	XGBoost	optical	session	max_depth=2; reg_lambda=20
con_012	XGBoost	optical	session	max_depth=4; reg_lambda=20
con_013	Random Forest	shape	session	min_samples_leaf=2; max_features=0.75
con_014	Random Forest	shape	session	min_samples_leaf=10; max_features=0.75
con_015	Random Forest	shape	session	min_samples_leaf=30; max_features=0.75
con_016	Extra Trees	shape	session	min_samples_leaf=3; max_features=0.75
con_017	Extra Trees	shape	session	min_samples_leaf=15; max_features=0.75
con_018	HistGB	shape	session	max_leaf_nodes=7; l2_regularization=10
con_019	HistGB	shape	session	max_leaf_nodes=15; l2_regularization=10
con_020	Logistic regression	shape	session	C=0.1
con_021	Logistic regression	shape	session	C=10
con_022	RBF SVC	shape	session	C=1; gamma_multiplier=1.0
con_023	RBF SVC	shape	session	C=10; gamma_multiplier=1.0
con_024	XGBoost	shape	session	max_depth=2; reg_lambda=20
con_025	XGBoost	shape	session	max_depth=4; reg_lambda=20
con_026	Random Forest	spatial	session	min_samples_leaf=2; max_features=0.75
con_027	Random Forest	spatial	session	min_samples_leaf=10; max_features=0.75
con_028	Random Forest	spatial	session	min_samples_leaf=30; max_features=0.75
con_029	Extra Trees	spatial	session	min_samples_leaf=3; max_features=0.75
con_030	Extra Trees	spatial	session	min_samples_leaf=15; max_features=0.75
con_031	HistGB	spatial	session	max_leaf_nodes=7; l2_regularization=10
con_032	HistGB	spatial	session	max_leaf_nodes=15; l2_regularization=10
con_033	Logistic regression	spatial	session	C=0.1
con_034	Logistic regression	spatial	session	C=10
con_035	RBF SVC	spatial	session	C=1; gamma_multiplier=1.0
con_036	RBF SVC	spatial	session	C=10; gamma_multiplier=1.0
con_037	XGBoost	spatial	session	max_depth=2; reg_lambda=20

ID	Family	Features	Weights	Varied parameters
con_038	XGBoost	spatial	session	max_depth=4; reg_lambda=20
con_039	Random Forest	temporal	session	min_samples_leaf=2; max_features=0.75
con_040	Random Forest	temporal	session	min_samples_leaf=10; max_features=0.75
con_041	Random Forest	temporal	session	min_samples_leaf=30; max_features=0.75
con_042	Extra Trees	temporal	session	min_samples_leaf=3; max_features=0.75
con_043	Extra Trees	temporal	session	min_samples_leaf=15; max_features=0.75
con_044	HistGB	temporal	session	max_leaf_nodes=7; l2_regularization=10
con_045	HistGB	temporal	session	max_leaf_nodes=15; l2_regularization=10
con_046	Logistic regression	temporal	session	C=0.1
con_047	Logistic regression	temporal	session	C=10
con_048	RBF SVC	temporal	session	C=1; gamma_multiplier=1.0
con_049	RBF SVC	temporal	session	C=10; gamma_multiplier=1.0
con_050	XGBoost	temporal	session	max_depth=2; reg_lambda=20
con_051	XGBoost	temporal	session	max_depth=4; reg_lambda=20

Appendix B. Exact estimator settings and saved components

The following constructor is the executed tuning factory. It combines with every candidate in Appendix A. Unspecified parameters follow the recorded library versions; complete get_params exports for the best full-pool components and the fixed benchmark are printed below. Serialized null values retain the source export representation, including the XGBoost missing-value default.

```
SEED = 1741
def estimator(config, n_features):
    task, family, p = config["task"], config["family"], dict(config["params"])
    regression = task == "force"
    if family in ("Extra Trees", "Random Forest"):
        cls = {("Extra Trees", True): ExtraTreesRegressor,
               ("Extra Trees", False): ExtraTreesClassifier,
               ("Random Forest", True): RandomForestRegressor,
               ("Random Forest", False): RandomForestClassifier}[family, regression]
        return cls(n_estimators=180, n_jobs=1, random_state=SEED, **p)
    if family == "HistGB":
        cls = HistGradientBoostingRegressor if regression else HistGradientBoostingClassifier
        return cls(max_iter=180, learning_rate=.05, min_samples_leaf=20,
                   early_stopping=False, random_state=SEED, **p)
    if family == "XGBoost":
        cls = XGBRegressor if regression else XGBClassifier
        return cls(n_estimators=220, learning_rate=.04, min_child_weight=5,
                   tree_method="hist", n_jobs=1, random_state=SEED, **p)
    if family == "Logistic regression":
        return LogisticRegression(max_iter=2500, random_state=SEED, **p)
    if family == "RBF SVR":
        return SVR(cache_size=256, **p)
    if family == "RBF SVC":
        p["gamma"] = p.pop("gamma_multiplier") / n_features
        return SVC(cache_size=256, decision_function_shape="ovr", random_state=SEED, **p)
    if family == "MLP":
        cls = MLPRegressor if regression else MLPClassifier
        return cls(max_iter=350, early_stopping=False, learning_rate_init=.001,
                   random_state=SEED, **p)
    raise ValueError(family)
```

Best-family full-pool components: complete parameter exports

force_for_056.joblib

categorical_features="from_dtype"; early_stopping=false; interaction_cst=null; l2_regularization=10; learning_rate=0.05; loss="squared_error"; max_bins=255; max_depth=null; max_features=1.0; max_iter=180; max_leaf_nodes=31; min_samples_leaf=20; monotonic_cst=null; n_iter_no_change=10; quantile=null; random_state=1741; scoring="loss"; tol=1e-07; validation_fraction=0.1; verbose=0; warm_start=false.

Scaler: with_mean=True; with_std=True. Feature order and dimension follow the named pack. Learned scaler arrays and estimator weights/trees are stored together in the joblib component.

SHA256: 338d605806bb042fbe47e53a1d0fd88b7bddc7f72cb7989a8d7b73aed9e853dc

force_for_050.joblib

bootstrap=false; ccp_alpha=0.0; criterion="squared_error"; max_depth=null; max_features=0.75; max_leaf_nodes=null; max_samples=null; min_impurity_decrease=0.0; min_samples_leaf=3; min_samples_split=2; min_weight_fraction_leaf=0.0; monotonic_cst=null; n_estimators=180; n_jobs=1; oob_score=false; random_state=1741; verbose=0; warm_start=false.

Scaler: with_mean=True; with_std=True. Feature order and dimension follow the named pack. Learned scaler arrays and estimator weights/trees are stored together in the joblib component.

SHA256: 6de9bc275c4578553b2111f461ccfa9b5468e28e0fbd07b389fa754e6b49b52e

force_for_059.joblib

base_score=null; booster=null; callbacks=null; colsample_bylevel=null; colsample_bynode=null; colsample_bytree=null; device=null; early_stopping_rounds=null; enable_categorical=true; eval_metric=null; feature_types=null; feature_weights=null; gamma=null; grow_policy=null; importance_type=null; interaction_constraints=null; learning_rate=0.04; max_bin=null; max_cat_threshold=null; max_cat_to_onehot=null; max_delta_step=null; max_depth=5; max_leaves=null; min_child_weight=5; missing="NaN"; monotone_constraints=null; multi_strategy=null; n_estimators=220; n_jobs=1; num_parallel_tree=null; objective="reg:squarederror"; random_state=1741; reg_alpha=null; reg_lambda=20; sampling_method=null; scale_pos_weight=null; subsample=null; tree_method="hist"; validate_parameters=null; verbosity=null.

Scaler: with_mean=True; with_std=True. Feature order and dimension follow the named pack. Learned scaler arrays and estimator weights/trees are stored together in the joblib component.

SHA256: 58044a24b1797c75daaf61cccc3e6e4cef2ebba3b92ea459ae8fab2f43846967

localization_loc_036.joblib

C=0.03; class_weight=null; dual=false; fit_intercept=true; intercept_scaling=1; l1_ratio=0.0; max_iter=2500; n_jobs=null; penalty="deprecated"; random_state=1741; solver="lbfgs"; tol=0.0001; verbose=0; warm_start=false.

Scaler: with_mean=True; with_std=True. Feature order and dimension follow the named pack. Learned scaler arrays and estimator weights/trees are stored together in the joblib component.

SHA256: 60afcde48ca3aa43c481619ff633b2e8571c477eed2d5404a282108e61fee95d

contact_con_022.joblib

C=1; break_ties=false; cache_size=256; class_weight=null; coef0=0.0; decision_function_shape="ovr"; degree=3; gamma=0.011111111111111112; kernel="rbf"; max_iter=-1; probability="deprecated"; random_state=1741; shrinking=true; tol=0.001; verbose=false.

Scaler: with_mean=True; with_std=True. Feature order and dimension follow the named pack. Learned scaler arrays and estimator weights/trees are stored together in the joblib component.

SHA256: d6d81520c54eb0f41cb4e18124e90500e1af0842d3ef884da2c931e7fd1b77de

For the TEST2 force member for_055, use the full HistGB parameter set above with max_leaf_nodes=15 instead of 31. For the TEST1 localization member loc_048, use the same Logistic-regression parameter set with temporal rather than invariant features and the fold-specific scaler/history. These are changes to the input pack and selected output history, not changes to C.

Fixed benchmark: complete parameters

Force estimators use positive=9. Total-light linear and isotonic reduce the scaled nine channels to their sum; the fitted median baseline ignores X. Classifiers use optical=27. Fixed classifiers use native decisions (probability threshold 0.5 or SVC decision threshold 0), with no output EMA or debounce. Fixed force predictions are not clipped. Contact and localization have the same recorded constructor parameters for each family; XGBoost's classifier resolves its multiclass objective during localization fitting. [S4, S7]

Force — Median baseline

constant=null; quantile=null; strategy="median".

Force — Total light linear

copy_X=true; fit_intercept=false; n_jobs=null; positive=true; tol=1e-06.

Force — NNLS

copy_X=true; fit_intercept=false; n_jobs=null; positive=true; tol=1e-06.

Force — Positive Ridge

alpha=10.0; copy_X=true; fit_intercept=false; max_iter=null; positive=true; random_state=null; solver="auto"; tol=0.0001.

Force — Isotonic

increasing=true; out_of_bounds="clip"; y_max=null; y_min=null.

Force — Monotonic HistGB

categorical_features="from_dtype"; early_stopping=false; interaction_cst=null; l2_regularization=10; learning_rate=0.05; loss="squared_error"; max_bins=255; max_depth=null; max_features=1.0; max_iter=150; max_leaf_nodes=7; min_samples_leaf=30; monotonic_cst=[1, 1, 1, 1, 1, 1, 1, 1, 1]; n_iter_no_change=10; quantile=null; random_state=1729; scoring="loss"; tol=1e-07; validation_fraction=0.1; verbose=0; warm_start=false.

Force — Monotonic XGBoost

base_score=null; booster=null; callbacks=null; colsample_bylevel=null; colsample_bynode=null; colsample_bytree=null; device=null; early_stopping_rounds=null; enable_categorical=true; eval_metric=null; feature_types=null; feature_weights=null; gamma=null; grow_policy=null; importance_type=null; interaction_constraints=null; learning_rate=0.05; max_bin=null; max_cat_threshold=null; max_cat_to_onehot=null; max_delta_step=null; max_depth=3; max_leaves=null; min_child_weight=10; missing=null; monotone_constraints="(1,1,1,1,1,1,1,1,1)"; multi_strategy=null; n_estimators=150; n_jobs=2; num_parallel_tree=null; objective="reg:squarederror"; random_state=1729; reg_alpha=null; reg_lambda=10; sampling_method=null; scale_pos_weight=null; subsample=null; tree_method="hist"; validate_parameters=null; verbosity=null.

Force — Extra Trees

bootstrap=false; ccp_alpha=0.0; criterion="squared_error"; max_depth=null; max_features=1.0; max_leaf_nodes=null; max_samples=null; min_impurity_decrease=0.0; min_samples_leaf=10; min_samples_split=2; min_weight_fraction_leaf=0.0; monotonic_cst=null; n_estimators=150; n_jobs=2; oob_score=false; random_state=1729; verbose=0; warm_start=false.

Force — RBF SVR

C=10; cache_size=200; coef0=0.0; degree=3; epsilon=0.1; gamma="scale"; kernel="rbf"; max_iter=-1; shrinking=true; tol=0.001; verbose=false.

Contact and localization — Logistic regression

C=1; class_weight=null; dual=false; fit_intercept=true; intercept_scaling=1; l1_ratio=0.0; max_iter=3000; n_jobs=null; penalty="deprecated"; random_state=1729; solver="lbfgs"; tol=0.0001; verbose=0; warm_start=false.

Contact and localization — RBF SVC

C=10; break_ties=false; cache_size=200; class_weight=null; coef0=0.0; decision_function_shape="ovr"; degree=3; gamma="scale"; kernel="rbf"; max_iter=-1; probability=false; random_state=1729; shrinking=true; tol=0.001; verbose=false.

Contact and localization — Random Forest

bootstrap=true; ccp_alpha=0.0; class_weight=null; criterion="gini"; max_depth=null; max_features="sqrt"; max_leaf_nodes=null; max_samples=null; min_impurity_decrease=0.0; min_samples_leaf=5; min_samples_split=2; min_weight_fraction_leaf=0.0; monotonic_cst=null; n_estimators=150; n_jobs=2; oob_score=false; random_state=1729; verbose=0; warm_start=false.

Contact and localization — Extra Trees

bootstrap=false; ccp_alpha=0.0; class_weight=null; criterion="gini"; max_depth=null; max_features=0.75; max_leaf_nodes=null; max_samples=null; min_impurity_decrease=0.0; min_samples_leaf=5; min_samples_split=2; min_weight_fraction_leaf=0.0; monotonic_cst=null; n_estimators=150; n_jobs=2; oob_score=false; random_state=1729; verbose=0; warm_start=false.

Contact and localization — HistGB

categorical_features="from_dtype"; class_weight=null; early_stopping=false; interaction_cst=null; l2_regularization=10; learning_rate=0.05; loss="log_loss"; max_bins=255; max_depth=null; max_features=1.0; max_iter=120; max_leaf_nodes=7; min_samples_leaf=25; monotonic_cst=null; n_iter_no_change=10; random_state=1729; scoring="loss"; tol=1e-07; validation_fraction=0.1; verbose=0; warm_start=false.

Contact and localization — XGBoost

base_score=null; booster=null; callbacks=null; colsample_bylevel=null; colsample_bynode=null; colsample_bytree=null; device=null; early_stopping_rounds=null; enable_categorical=true; eval_metric=null; feature_types=null; feature_weights=null; gamma=null; grow_policy=null; importance_type=null; interaction_constraints=null; learning_rate=0.05; max_bin=null; max_cat_threshold=null; max_cat_to_onehot=null; max_delta_step=null; max_depth=3; max_leaves=null; min_child_weight=5; missing=null; monotone_constraints=null; multi_strategy=null; n_estimators=120; n_jobs=2; num_parallel_tree=null; objective="binary:logistic"; random_state=1729; reg_alpha=null; reg_lambda=10; sampling_method=null; scale_pos_weight=null; subsample=null; tree_method="hist"; validate_parameters=null; verbosity=null.

Appendix C. Exact feature-construction source

The following function preserves the executed block order. AREA_COLUMNS, RAW_LIGHT_COLUMNS, SIGNED and ACTIVE_COLUMNS list the nine taxels in ascending order; they respectively contain ROI areas, raw positive-light sums, signed-light sums and active fractions. Frames must already be ordered chronologically within recording. The input floor is the training-only nine-vector described in Section 3.

```
def causal_ema(values, session_ids, times, tau):
    """Only current/past rows; restart at session boundary or >0.3 s gap."""
    if tau == 0:
        return np.asarray(values).copy()
    values = np.asarray(values)
    result = np.empty_like(values, dtype=float)
    for i in range(len(values)):
        dt = times[i] - times[i-1] if i else 0
        if i == 0 or session_ids[i] != session_ids[i-1] or dt > 0.3 or dt <= 0:
            result[i] = values[i]
        else:
            a = -np.expm1(-dt / tau)
            result[i] = result[i-1] + a * (values[i] - result[i-1])
    return result


def optical_features(frames, floors):
    """Return fixed optical feature packs; labels/force/absolute time are absent."""
    def cols(s): return frames[[f"roi{r}_{s}" for r in range(1,10)]].to_numpy(float)
    area = frames[AREA_COLUMNS].to_numpy(float)
    if not (np.isfinite(area).all() and (area > 0).all()):
        raise ValueError("Invalid ROI areas")
    positive = np.maximum(frames[RAW_LIGHT_COLUMNS].to_numpy(float) - floors, 0) / area
    signed = frames[SIGNED].to_numpy(float) / area
    active = frames[ACTIVE_COLUMNS].to_numpy(float)
    optical = np.column_stack([positive, signed, active])
    if not np.isfinite(optical).all(): raise ValueError("Nonfinite basic optical features")
    def unit(x): return x / (np.sum(np.abs(x), axis=1, keepdims=True) + 1e-6)
    def contrast(x): return x - np.median(x, axis=1, keepdims=True)
    med, mad = cols("signed_delta_v_median"), cols("signed_delta_v_mad")
    raw = cols("raw_v_mean")
    rawmed, rawmax = cols("raw_v_median"), cols("raw_v_max")


    cx = (cols("centroid_x") - cols("x")) / np.maximum(cols("width"), 1)
    cy = (cols("centroid_y") - cols("y")) / np.maximum(cols("height"), 1)
    present = (np.isfinite(cx) & np.isfinite(cy)).astype(float)
    cx, cy = np.nan_to_num(cx, nan=0.5), np.nan_to_num(cy, nan=0.5)
    shape = np.column_stack([unit(positive), unit(contrast(signed)), unit(active),
        contrast(signed), contrast(active), unit(contrast(raw)), unit(contrast(med))])
    spatial = np.column_stack([optical, shape, med, mad, cx, cy, present,
        contrast(raw), contrast(rawmed), rawmax / (raw + 1),
        np.log1p(positive), np.sign(signed) * np.log1p(np.abs(signed))])
    invariant = np.column_stack([shape, unit(mad), cx, cy, present,
        unit(contrast(rawmed)), rawmax / (raw + 1)])
    ids = frames.session_id.to_numpy()
    times = frames.capture_monotonic_relative_s.to_numpy(float)
    fast = causal_ema(optical, ids, times, .3)
    slow = causal_ema(optical, ids, times, 1.0)
    temporal = np.column_stack([spatial, fast, slow, optical-fast, fast-slow])
    packs = dict(positive=positive, optical=optical,
        shape=np.column_stack([optical, shape]), spatial=spatial,
        invariant=invariant, temporal=temporal)
    for name, x in packs.items():
        if not np.isfinite(x).all(): raise ValueError(f"Nonfinite {name} features")
    return packs
```

The feature code computes histories for all available optical frames before eligibility masking. Neither the target force nor target taxel enters the estimator feature array. No neural-network segmentation, image morphology, future-frame filtering or zero-response rejection is applied.

Appendix D. Evidence register and artifact integrity

Paths below are relative to the shared SOFTWARE CALIBRATION project unless noted. They identify inspectable project evidence, not external academic literature. This PDF embeds the numbers and figures needed for thesis writing; the source files remain available for deeper replication.

S1 — Consolidated best metrics, matrices and fold configurations

reports/BEST_RESULTS/BEST_METRICS.json

SHA256: 0f74601199ca2b644e9abc2f83ab402f516a37d99c1da70728616f564e754332

S2 — All 50 fixed and tuned comparison rows

reports/BEST_RESULTS/reference/all_competing_models_ranked.csv

SHA256: 0a4e85fb7d19ae7b50601480951f5484c56876d2fe613ad238f4f16e13326370

S3 — Winning-model held-out prediction rows

reports/BEST_RESULTS/data/best_models_held_out_predictions.parquet

SHA256: 84ba96166137f8ab1c71c6c8d9303b1e75297bfc18e5e89f55e2e373596b9560

S4 — Fixed benchmark protocol

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/retraining_20260909/protocol.json

SHA256: 31924ffc9ba96124719f2fe4f4e057492716827a0ef4bbbde995f501b5b9f28a

S5 — Tuning protocol and full 178-candidate grid

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/tuning_20260909/protocol.json

SHA256: 81ecf501fb217d40f33dc352da1fe123f6a888c4f0a2effad6398375488a523e

S6 — Fixed-run dataset and feature provenance

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/retraining_20260909/provenance.json

SHA256: f0fa7fdc78bdaae23dd9b1fc2bd47c21530b20f0c5389c8c7b7392429319f3af

S7 — Fixed estimator parameter export

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/retraining_20260909/estimator_parameters.json

SHA256: df26dd967959929a56b3b98100c1dc007e2f09e45d269902434ea497b3cf5c75

S8 — Exact segmentation, model defaults and feature source

reports/manual_calibration_thesis_report_2026-10-05/Exact_Segmentation_and_Model_Configurations.json

SHA256: f11515378af507a6d7b064c257d84a37390608180858609876 32e7de62750a15

S9 — Tuning validation audit

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/tuning_20260909/validation_audit.json

SHA256: e1400aa5a1b91b788751c6ff217e88647594e1a829c80aafa383080fd03db3a5

S10 — Optical source validation

reports/manual_calibration_matrices_and_light_metrics_2026-10-18/source_validation.json

SHA256: abac5f5948e7e2f278bf3942e7eae5c2530e3db695c9f887d8b7381843ddb285

S11 — Optical analysis methods and interpretation

reports/manual_calibration_matrices_and_light_metrics_2026-10-18/Technical_Notes_for_Thesis.txt

SHA256: 48ae27c36b1a6eca4415b36213bf50ec535872adf0ea07b3b6f3985c76ac210a

S12 — Optical metric calculations and validation

reports/manual_calibration_matrices_and_light_metrics_2026-10-18/metrics_and_validation.json

SHA256: f65eb208b539728ccca25747df1e41c6abf4272759727d190de6a3b03a15abe0

S13 — Independent winner-metric and matrix recomputation for this report

reports/complete_thesis_report_2026-10-18/derived_evidence.json

SHA256: cb7514a3dc2432b2a1cdbb871fecff843fda36533bdc17aaef9cd3685dbfb8c0

S14 — Saved original-bundle timing audit

calibration_gui/analysis_outputs/live_sensor_manual_only/model_runs/tuning_20260909/inference_timing.json

SHA256: 5de2b3fd743678edb665916ce290188fce8a1cb8587e06a781f5c78a6f97f1c0

S15 — Prior report: acquisition context and hardware evidence register

reports/manual_calibration_thesis_report_2026-10-05/Manual_Calibration_Models_Technical_Report.txt

SHA256: 29720f548ca9fec0a1523da485e902d0f8da144480d3d3733ce6e8a09828cc8d

S16 — Direct HSV V metrics, baseline provenance and calculation checks

reports/manual_v_channel_metrics_2026-10-18/v_channel_metrics.json

SHA256: 8c44db5108c12856b8df0d213eb4dacbe9dd84f49d2694fd2ebf5fe47f49408b

Manual archive

External original path: C:/Users/DLSU/Downloads/THESIS DATA/Data Collection/Manual Calibration (2).zip. Recorded size: 1,283,430,583 bytes. SHA256: e039dcd49f3390d34dc979490dcf46a6ecc43d9e6a2aec08e4952acc14b5006a. The report does not import automatic-calibration data.

Additional reproducibility files

BEST_RESULTS/BEST_MODEL_CONFIGURATIONS.json provides all six winning-family fold settings; BEST_RESULTS/figure_manifest.json and Figure_Captions.txt map the eight reused figures. The tuning run's selections_fold1.json through selections_fold6.json, final_selections.json, source_snapshot/, environment.json and models/ preserve the detailed run. The raw manual optical source-member manifest and all light CSV tables remain under BEST_RESULTS/data/.

The complete comparison table contains all 50 evaluated family/procedure rows. The candidate appendix contains all 178 configurations. Every matrix and quantitative heatmap has its numeric values printed in searchable tables. No competing model, previous report, prediction row, or source figure is removed by this update.